

The Missing Healthcare Opportunity

*Healthcare Has More Information Than Ever.
Why Does It Still **Cost So Much**?*



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Introduction

The proposal in this book is **audacious**.

It suggests that the United States should consider creating a national information architecture capable of maintaining a trustworthy longitudinal history for each participating patient—drawing information from healthcare organizations that today operate independently across a fragmented healthcare system.

That history could encompass decades of diagnoses, clinical notes, laboratory results, imaging, medications, procedures, treatments, claims, authorizations, and outcomes. Information would have to be matched to the correct patient, reconciled when sources disagree, preserved with its provenance, protected from unauthorized use, and made available in the context required for a particular decision.

Artificial intelligence would play two important roles.

1. After a patient encounter or new result, AI could help extract clinical meaning, reconcile it with the existing history, preserve unresolved questions, capture outcomes, and integrate the new information into a trustworthy longitudinal patient history while preserving links to the underlying evidence
2. When a clinical event occurs, AI could retrieve and summarize the relevant knowledge—including unresolved questions, risks, gaps, and prior outcomes—and relate it to treatments and outcomes for similar patients. It could also incorporate relevant clinical guidelines, research, and other external medical knowledge.

The federal government is moving in this direction. TEFCA, the CMS Health Technology Ecosystem, and CDC data-modernization initiatives are expanding interoperability, patient access, and the ability to transform healthcare and public-health data into usable information. These efforts are making healthcare information more accessible, **but access is not the same as integration.** They do not yet provide the continuously maintained, evidence-linked longitudinal patient history proposed in this book.

The objective is not to replace the judgment of physicians, payers, or other healthcare professionals. It is to give them better information on which to exercise that judgment.

- A physician should be able to see what has already been tried, what is known about the outcome, what remains unresolved, what risks or changes deserve attention, and what approaches have produced better outcomes for similar patients.
- A payer considering an authorization should have the clinical context needed to understand why a treatment is being requested, along with evidence on the costs and outcomes of alternative approaches.

The goal is better-informed decisions: the right care at the right time, with less delay, duplication, and administrative friction.

This is not only a clinical information problem. Healthcare is also commerce. Patients choose providers and treatments. Providers recommend and deliver care. Payers authorize and pay for it. Yet these decisions are often made without timely information about availability, experience, cost, alternatives, and outcomes.

The potential economic opportunity is enormous. U.S. healthcare spending is measured in trillions of dollars. Administrative costs, improper payments, repeated work, delayed decisions, and care delivered without relevant history create opportunities measured in billions. **Even relatively small improvements could produce substantial economic value.**

But the scale of the opportunity is not evidence that this proposal is feasible.

Creating such an infrastructure raises substantial technical, economic, privacy, legal, and political questions. This book does not claim to have resolved them or recommend beginning with a massive government technology program.

It recommends something much smaller: **determine whether the idea is worth pursuing.**

The chapters that follow examine the problem, the missing information architecture, how a trustworthy longitudinal patient history could be created and used, and its potential benefits. The book concludes with a six-person, 60-day feasibility assessment to determine whether the idea warrants further investigation.

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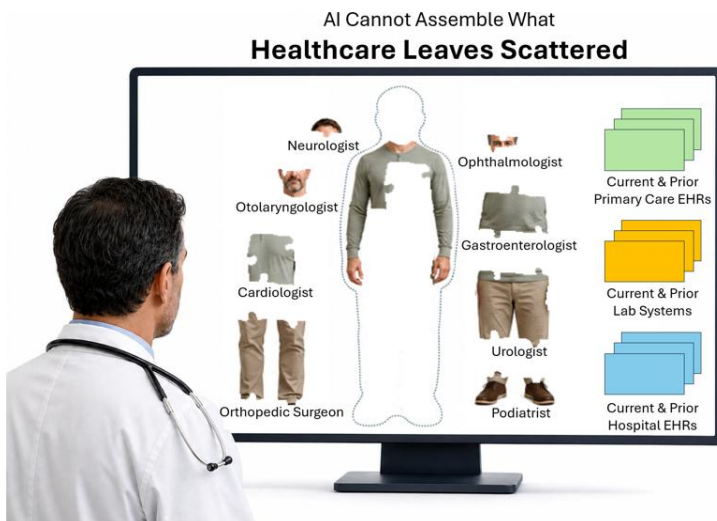
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Missing the Macro Opportunity in Healthcare

Healthcare has evolved from care often centered on a family physician familiar with much of a patient's history to care delivered by an increasingly specialized network of primary-care physicians, specialists, urgent-care centers, hospitals, laboratories, imaging centers, pharmacies, and other providers. **The expertise has become more specialized, but the patient's history has become more fragmented.** One provider may recommend a treatment, another deliver it, and the outcome may never be systematically captured or shared with the rest of the patient's healthcare team.

This chapter presents the book's central vision: a secure national longitudinal patient history that could bring that fragmented information together, reduce administrative costs, and give healthcare professionals more complete information for better decisions. AI could help transform it into useful clinical context—supporting, rather than replacing human judgment.



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Background

Micro Advances, Macro Opportunity

We regularly hear about narrowly targeted uses of artificial intelligence in healthcare.

AI models can help identify suspicious areas in medical images, compare current scans with earlier ones, and draw a physician's attention to changes that might otherwise be missed. AI can help clinicians prepare more complete notes while spending less time typing and more time with patients.

These can be important—even life-changing—advances. But they apply AI to individual tasks within the healthcare system. They do little to address fragmented information and administrative friction that affect the entire system.

In fact, a micro improvement for one participant does not necessarily reduce the total cost of healthcare. More complete clinical documentation may justify more accurate—and sometimes higher—reimbursement. A provider operating under negotiated payment rates may benefit from greater productivity, but those savings may not reach the insurer or the patient. Claims of savings may also not include the full costs of development, integration, infrastructure, maintenance, and oversight.

A recent systematic review found promising economic outcomes from clinical AI applications, including fewer unnecessary procedures and more efficient use of resources. But it examined only 19 studies across selected applications and called for more standardized, long-term economic evaluation. That is evidence that some AI applications can be cost-effective—not yet evidence that adding AI throughout today's healthcare system will reduce its total cost.¹

Meanwhile, a much larger opportunity to reduce costs and improve effectiveness is sitting in front of us... Sometimes literally.

¹ El Arab, Rabie Adel, and Omayma Abdulaziz Al Moosa. "Systematic Review of Cost Effectiveness and Budget Impact of Artificial Intelligence in Healthcare." *npj Digital Medicine* 8 (2025): 548. DOI: 10.1038/s41746-025-01722-y. Review of 19 studies found generally favorable clinical and economic results for AI applications, while noting incomplete accounting for implementation and indirect costs and limitations in long-term economic modeling.

Part of the Cost Was Sitting in My Lap

I turned 76 this year. Since moving to Florida, I have had four primary-care physicians. This year, my newest one referred me to a cardiologist, an orthopedic specialist, and a physical therapist.

At each office, I was handed a clipboard and asked for nearly the same information.

Once again, I tried to remember the dates of surgeries and injuries. Once again, I worked through pages of medical conditions, checking yes or no. Much of the information already existed in the records of my previous physicians—but apparently “somewhere in my medical history” was not the same as “available here.”

While I completed one set of forms, the waiting-room television carried another news story about the rising cost of healthcare.

CRITICAL INFORMATION		
Adhesive tape / latex allergy	☐ Yes	☐ No
Anticoagulant treatment	☐ Yes	☐ No
Artificial heart valves	☐ Yes	☐ No
Artificial joint	☐ Yes	☐ No
Bacitracin / Neosporin allergy	☐ Yes	☐ No
Bleeding disorders	☐ Yes	☐ No
Breast cancer / Other cancer	☐ Yes	☐ No
Epilepsy	☐ Yes	☐ No
Epinephrine sensitivity	☐ Yes	☐ No
...		

I looked at the pages on my clipboard and wondered who would enter this information, whether anyone would reconcile it with my other records, and what would happen if my 76-year-old memory supplied the wrong answer.

Then I looked beyond the receptionist at the people working on scheduling, insurance verification, claims, prior authorizations, payments, and medical records for a practice with two physicians.

As the television continued asking why healthcare costs so much, I began to suspect that part of the answer was sitting in my lap.

While AI contributes to micro improvements in healthcare, it could also attack enormous macro-opportunities:

- reduce provider–payer administrative effort;
- prevent unnecessary duplication of tests and treatments;
- use complete longitudinal information to help clinicians make more accurate decisions and keep people healthier.

We are using AI to improve individual healthcare tasks while overlooking the information architecture needed to improve healthcare as a system.

Why I Am Raising the Question

I am not a doctor, and I don't play one on TV. I do claim considerable expertise in determining whether an information architecture can deliver the information required to support better decisions and outcomes.

Over my career, I participated in four-to-eight-week benchmark engagements with more than 100 major corporations across many industries, including several involved in healthcare. The technology varied, but the fundamental question did not:

Could the organization get the right information to the right person in time to make a better decision?

In healthcare, that question can affect both cost and survival.

When Existing Information Is Not Available

The two personal examples establish two different failures:

- The accumulated history exists but is too fragmented to guide continuing treatment.
- A specific technical fact exists but is unavailable at the moment of treatment.

A friend of mine has been battling disabling intestinal problems for nearly four years. Different specialists have ordered scans and colonoscopies, prescribed medications, and recommended specialized physical therapy. Yet the periodic episodes continue.

Were the previous tests, treatments, and results available to the current specialist in a form that could be readily understood? Had the healthcare system learned from each unsuccessful intervention—or merely recorded it somewhere? Could the experience of specialists treating comparable patients offer another course of action?

When my stepson was receiving chemotherapy through a port in his chest, the staff preparing an injection asked him what size needle had been used previously!!

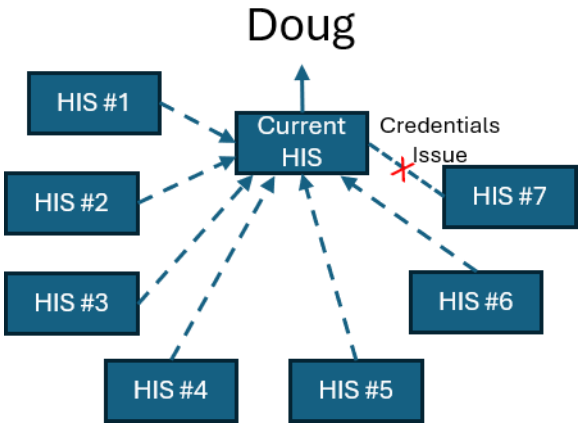
Why was the patient being asked to supply a technical detail needed to treat him safely? What could happen if he gave the wrong answer?

The information existed. It simply was not available when it was needed.

More Data, Less Usable History

Our advanced technology is generating more health data than ever. At the same time, increasing specialization is spreading that information across more providers and systems. Each specialist may successfully treat the part of the patient assigned to them while no one has a complete view of the patient.

When I began seeing my new primary-care provider this year, I updated my information in the practice's **Health Information System (HIS)**. I linked six other healthcare information systems to my account. I couldn't link a seventh because of issues with my login credentials.



Connected - But not integrated

That does not include records from three dental providers and two dental specialists I have used during my adult life.

Technically, I have created pathways to much of my history. That does not mean it has been integrated, reconciled, or made useful.

For a new provider to understand my history, someone would have to search years of physician notes, test results, diagnoses, medications, and procedures distributed across those systems. Identifying the information relevant to a new condition could require an hour or more of searching and study—time most providers do not have.

This is not a modern information architecture designed for people who change jobs, move to new cities, and receive care from multiple primary-care providers and specialists.

It does not create a reliable longitudinal history in which identities are matched, duplicate records are recognized, conflicting information is identified, and important facts are validated.

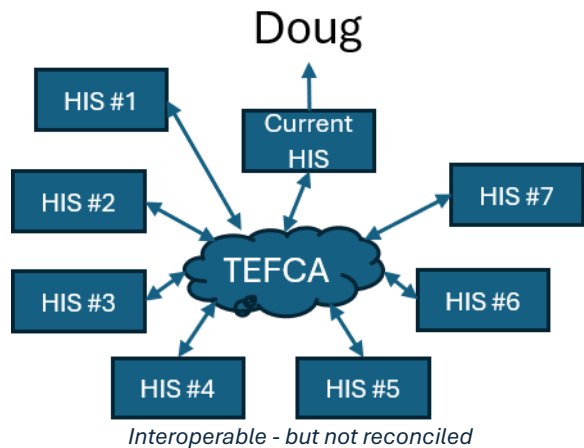
Without that foundation, AI cannot reliably extract the history relevant to a current decision, identify potentially redundant tests and treatments, compare results over time, improve authorization decisions, or evaluate costs and risk-adjusted outcomes across providers.

AI does not replace healthcare providers' judgment. It can do the work no provider has time for: continuously examining years of distributed information and presenting relevant evidence when decisions are made about diagnoses, treatments, referrals, and prescriptions.

TEFCA: Important Progress, Not the Complete Foundation

The federal government and healthcare industry have worked for decades to improve healthcare information exchange. Early federal efforts such as the Nationwide Health Information Network (NHIN) and standards and profiles developed through Integrating the Healthcare Enterprise (IHE) established many of the capabilities needed to locate, exchange, and securely access patient information.

TEFCA²—the Trusted Exchange Framework and Common Agreement—builds on that progress with a nationwide framework through which participating healthcare organizations can query and exchange patient information. As participation expands, TEFCA may eliminate some of the portal linking, passwords, and manual record transfers that patients experience today.



² Office of the National Coordinator for Health Information Technology (ONC). "Advancing Nationwide Interoperability with TEFCA."

<https://healthit.gov/policy/tefca/>. TEFCA establishes a nationwide framework for electronic health-information sharing, enabling providers, payers, public-health professionals, and patients to securely access and exchange information across participating networks.

That is important progress. But exchanging records is not the same as integrating the information they contain.

TEFCA does not, by itself, create a continuously maintained longitudinal history in which records from different sources have been matched, duplicate events identified, conflicting information reconciled, and outcomes connected to treatments. It does not create the analytical foundation needed to compare costs and risk-adjusted outcomes across providers. Nor does it prepare the relevant facts for a physician before a treatment decision is made.

TEFCA could provide an important channel for information to reach such a foundation. It is not the complete foundation.

The macro opportunity requires an information architecture organized around the patient—not the institutions that hold fragments of the patient's history.

Recommendations:

1. Create a trusted longitudinal health history organized around each patient.
2. Use national standards to collect clinical, claims, laboratory, imaging, pharmacy, and other relevant information.
3. Preserve the source of every fact while identifying duplicates, conflicts, and missing information.
4. Use AI to present relevant history and evidence to providers at the point of decision.
5. Automate routine information exchange between providers and payers.
6. Use risk-adjusted outcomes to compare treatments and providers serving comparable patients.
7. Give patients control, visibility, and a complete audit trail of access to their information.
8. Permit approved analysis of protected data without exposing individual identities unnecessarily.
9. Use TEFCA and existing healthcare systems as information sources rather than expecting them to perform the integration.
10. Measure success by reductions in administrative effort, avoidable care and adverse outcomes—not by the volume of data exchanged.

Discussion:

Benefits to the Individual:

The individual—not the provider, payer, or government—should be the principal beneficiary of a patient-centered information architecture.

Today, patients repeatedly supply the same information, search for records, remember prior treatments, and carry technical details from one provider to another. Yet they generally cannot see a complete history, determine who has examined it, or easily compare results from different providers.

A Usable, Patient-Controlled History

A longitudinal health information architecture would give individuals:

- **One understandable health history.** Patients could see diagnoses, medications, procedures, test results, implanted devices, and significant events collected across providers.
- **Visibility into access.** Patients could see who accessed their information, when it was accessed, what information was retrieved, and the stated purpose.
- **Fewer unnecessary forms.** Patients would not have to reconstruct the same history from memory whenever they changed providers or are referred to specialists.
- **Fewer avoidable tests and scans.** Providers could see prior tests, retrieve the results, and determine whether repetition was clinically necessary.
- **Safer, better-informed treatment.** AI could identify relevant allergies, prior complications, unsuccessful treatments, medication conflicts, and changes in test results for provider review.
- **Faster, better-informed prior-authorization decisions.** Relevant clinical evidence could automatically accompany requests, reducing denials and resubmissions due to incomplete information.
- **Continuity when moving or changing providers.** A patient's usable history would not remain trapped in the city, health system, or employer they left behind.

- **Better information for selecting providers.** Patients could compare costs, experiences, and risk-adjusted outcomes across providers treating people with comparable conditions.
- **A clearer understanding of treatment choices.** Cost information could be considered alongside outcomes, complications, recovery times, and the experience of similar patients.
- **The ability to identify errors and conflicts.** Patients could flag incorrect information and see how providers resolved disputed or inconsistent entries.

Making Patient Choice Meaningful

The least expensive treatment is not cost-effective if it fails, causes complications, or requires repeat treatment. Patients need to compare the total cost of care with the outcomes achieved.

Patients are routinely told to become informed consumers of healthcare, but they are denied much of the information required to do so. Healthcare often does not systematically capture what happened after treatment, and physicians and patients cannot readily examine outcomes across people with similar conditions and characteristics. A patient's own history is also scattered across multiple systems.

Treatment choices can be even harder. A physician may recommend a painful treatment with significant side effects, while the Internet offers seemingly easier alternatives supported by testimonials. But improvement after a treatment does not prove the treatment caused it. The improvement claimed by the testimonial may have been coincidental to the alternative treatment and not a direct result. What is missing is evidence showing how comparable patients did with different treatments.

Patient choice becomes meaningful only when patients have usable information.

Better-Informed Prior Authorization

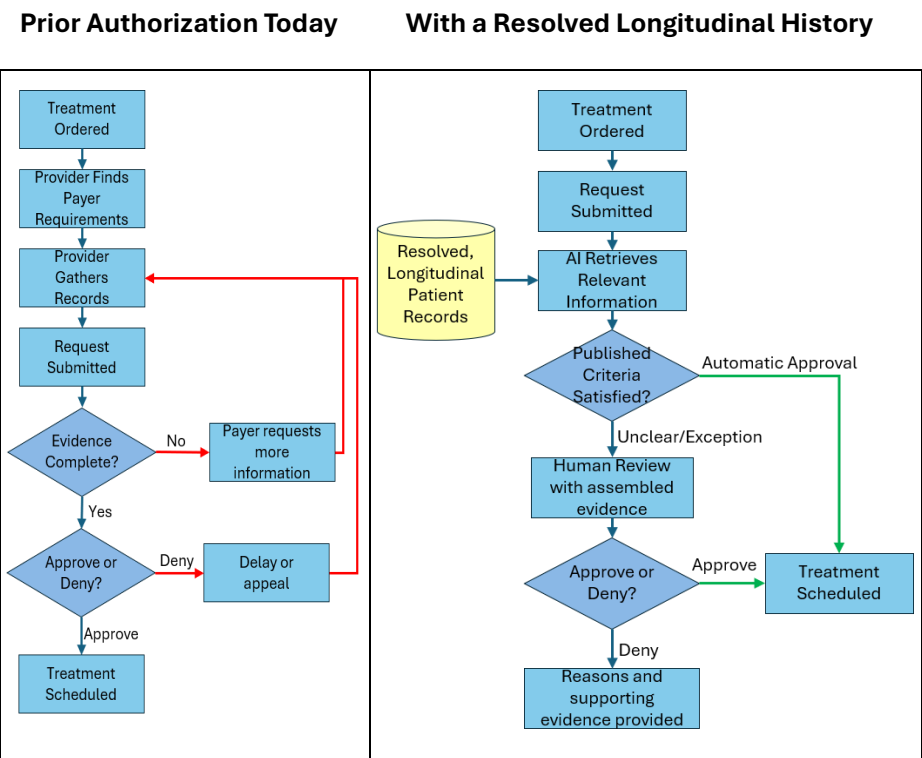
A significant pain point for individuals is prior authorization denial and the resulting delay in service.

Prior authorization is intended to prevent unnecessary or inappropriate treatment. In practice, a decision may be made without a complete understanding of the patient’s history.

A patient may be denied an MRI and required to complete weeks or months of physical therapy first. That may be appropriate for some patients. It may be inappropriate for a patient whose history, symptoms, previous treatment, or risk factors indicate the need for imaging.

Today, the provider may need to assemble and resubmit the evidence manually. The patient waits while the provider and payer exchange forms, notes, and telephone calls.

A longitudinal information architecture could automatically provide the payer with the information necessary for the decision: prior physical therapy, medications, earlier imaging, relevant diagnoses, symptom progression, and the results of similar treatment decisions involving comparable patients.



Better information should produce both more appropriate approvals and more defensible denials. The objective is not to approve every request. It is to reduce decisions made from incomplete information.

Benefits to the Provider:

Healthcare providers do not suffer from a shortage of patient data; they suffer from insufficient time to find, reconcile, and review it. Relevant information may be buried in years of physician notes, test results, images, prescriptions, and records maintained by other organizations.

A resolved longitudinal history would allow AI to identify pertinent conditions, prior treatments, medication changes, and trends before a clinical decision is made. The same architecture could reduce administrative work related to authorizations, claims, referrals, and record exchange, allowing clinicians and their office staff to spend more time providing care.

Relevant Information at the Point of Care

- relevant patient history presented before the encounter;
- trends and meaningful changes are highlighted automatically;
- fewer hours spent searching external records and reading repetitive notes;
- identification of prior unsuccessful treatments, complications and contraindications;
- automatic comparison of current and previous tests and images;
- source links allowing providers to verify every AI-generated summary;
- more complete referrals and transitions between providers;
- automatically prepared prior-authorization documentation;
- fewer claim rejections caused by incomplete or inconsistent information;
- less manual handling of records, forms, coding, and payer inquiries;
- more time for patients and clinical judgment.

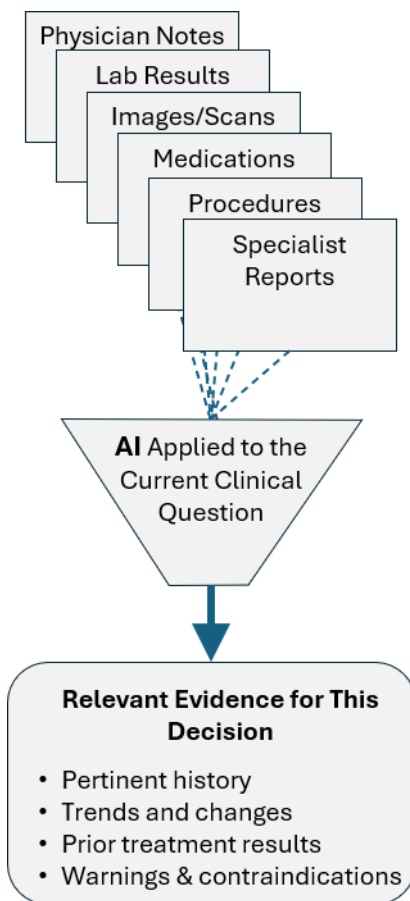
Making the Accumulated Record Usable

Healthcare providers face a paradox: they have access to more patient data than ever, yet less practical ability to understand it all before making a decision.

An older patient with several chronic conditions may have decades of physician notes, laboratory results, scans, procedures, medications, and specialist reports. When those records are distributed across multiple systems, a provider may not even know what exists. When records are available, the volume can make comprehensive review impossible within the time allowed for an appointment.

In practice, the most consistently visible information is often the current medication list, allergies, active diagnoses, and recent results. Important background can remain buried in years of repetitive notes and disconnected test results.

A study of outpatient notes found that the notes increased by 60% in length between 2009 and 2018. By 2018, approximately 71% of the text was templated or copied rather than directly entered. Longer records therefore do not necessarily provide proportionately more useful information. They can make the important facts harder to find³.



³ Rule, Adam, Steven Bedrick, Michael F. Chiang, and Michelle R. Hribar. "Length and Redundancy of Outpatient Progress Notes Across a Decade at an Academic Medical Center." *JAMA Network Open* 4, no. 7 (2021): e2115334. DOI: 10.1001/jamanetworkopen.2021.15334. Analysis of nearly 3 million outpatient notes found that median note length increased 60.1% from 2009 to 2018; in 2018, 70.6% of note text was templated or copied rather than directly typed.

Reviews of electronic-record information overload have identified potential consequences including missed information, delayed care, and patient-safety risks.⁴

AI documentation tools may help clinicians produce notes more efficiently. They may also create more detailed notes, adding still more information for the next provider to review.

AI should not only help create the record. It should help make the accumulated record usable.

Reducing Medical-Office Administration

The same longitudinal history could reduce work throughout the medical office. Staff would not have to collect demographic and historical information repeatedly. Referrals could automatically include the relevant history. Prior-authorization requests could be assembled with the diagnoses, earlier treatments, and test results required by the payer. Claims could be checked for completeness and consistency before submission. Denials could be matched with the evidence necessary for review or appeal.

This would not eliminate every administrative position. It would eliminate much of the searching, rekeying, reconciling, faxing, telephoning, and resubmitting that consumes staff time without directly improving patient care.

Providers do not need more data presented to them. They need the relevant evidence extracted from the data they already have—and from the data they currently cannot see.

⁴ Nijor, S., G. Rallis, N. Lad, et al. “Patient Safety Issues from Information Overload in Electronic Medical Records.” *Journal of Patient Safety* 18, no. 6 (2022): e999–e1003. DOI: 10.1097/PTS.0000000000001002. A literature review found that information overload associated with electronic health records can contribute to errors and threaten patient safety. AHRQ Patient Safety Network: <https://psnet.ahrq.gov/issue/patient-safety-issues-information-overload-electronic-medical-records>.

Benefits to the Payer:

A longitudinal health information architecture would provide payers with timely, complete evidence for authorization and payment decisions, reduce administrative effort and improper or fraudulent claims, and help determine whether coverage policies reduce total costs while improving patient outcomes.

Better Information for Decisions

A longitudinal health information architecture could provide:

- More complete and accurate claims.
- Less manual review and reconciliation.
- Fewer requests for missing documentation.
- Faster, more consistent prior-authorization decisions.
- Fewer avoidable appeals and overturned denials.
- Better coordination when members change insurers.
- Improved detection of duplicate, inconsistent, or potentially fraudulent claims.
- Better identification of treatments producing the best outcomes for comparable patients.
- Risk-adjusted comparisons of provider cost and performance.
- Earlier identification of patients who may benefit from preventive or coordinated care.
- Evidence showing whether coverage policies reduce total cost or merely postpone it.

Payers possess broad claims histories, but claims primarily describe what was billed and paid. They frequently lack the clinical detail needed to understand why a service was requested, what had already been attempted, and what happened to the patient afterward.

Prior Authorizations • Claims • Payments

Limits of the Payer’s Information
OFTEN HIDDEN OR DISCONNECTED

Clinical Context • Prior Treatment Results
Patient Outcomes • Costs Under Other Payers
Later Complications • Total Cost Over Time

That missing information creates work on both sides of the provider–payer interface. Payers request documentation, provider offices locate and submit it, and payers review it. Missing information leads to another request, a denial, or an appeal. The same evidence may be assembled and reviewed repeatedly by different people in different organizations.

A resolved longitudinal history could supply the relevant evidence electronically and in a consistent form. Claims could be checked against the documented diagnosis, treatment, and authorization before submission. Missing or conflicting information could be identified immediately rather than after the claim is rejected.

Prior authorization should become an evidence-based decision rather than a document-collection contest.

AI could assemble the symptoms, diagnoses, prior treatments, results, adverse effects, and applicable coverage rules. Straightforward requests could be processed consistently, while unusual, conflicting, or high-risk cases would be directed to qualified reviewers with the relevant evidence already assembled.

The standard should not be “AI decides.” AI should assemble the evidence and apply explicit rules. Accountable people should remain responsible for exceptional or disputed decisions.

Measuring Total Cost and Outcomes

A payer can calculate how much was saved by denying an MRI today. It may have much less visibility into whether the patient later required emergency treatment, surgery, hospitalization, or months of ineffective therapy.

A longitudinal history could connect an authorization decision with the eventual cost and outcome—even if the patient changed providers or insurers. Payers could then distinguish between unnecessary care, appropriately delayed care, and care whose delay increased total cost or contributed to a worse outcome. They could also identify treatments that cost more initially but reduce subsequent complications.

This addresses a structural problem created when people change employers and insurance plans. One payer may bear the cost of preventive treatment while another receives the future savings. Conversely, delaying treatment

may reduce one payer's immediate expense while leaving a future payer, Medicare, or the patient with a higher cost.

Healthcare should not declare a decision cost-effective merely because its consequences occurred after the patient changed insurance companies.

Payment Integrity and Accountability

Claims and clinical evidence could be compared to identify duplicate billing, services unsupported by the clinical record, incompatible procedures, and unusual patterns across providers and payers. Such findings should trigger review rather than automatic accusations. An apparent inconsistency could result from fraud, coding error, incomplete information, or legitimate clinical circumstances.

Payers would receive better information and lower administrative costs, but they would also become more accountable. Approval times, denial rates, appeal reversals, repeated requests for information, and outcomes following delayed or denied treatment could be compared across payers.

Payers should welcome better information—but better information must create accountability on both sides of the transaction.

How Current Healthcare Information Systems Would Change

The United States has invested enormous amounts of money and effort in healthcare information systems. Replacing them would be financially unrealistic, operationally disruptive, and unnecessary.

These systems perform many functions, which are best managed close to the organization delivering care. Physician practices need systems that support appointments, clinical documentation, prescriptions, referrals, orders, billing, and patient communication. Hospitals need systems that support admissions, beds, laboratories, imaging, pharmacies, and thousands of specialized workflows. Payers need systems that support enrollment, benefits, claims, authorization, and payment.

Those responsibilities would remain.

What would change is the expectation that each system independently maintains complete patient history.

A clearer division of responsibility

Existing healthcare systems	National architecture
Collect information during care	Assemble the longitudinal history
Support local clinical workflows	Match information to the correct patient
Schedule appointments and resources	Reconcile duplicate and conflicting information
Enter orders and prescriptions	Preserve provenance across sources
Capture physician and nursing notes	Identify clinically relevant history
Operate laboratories and imaging	Analyze trends across organizations and time
Submit and process claims	Compare risk-adjusted costs and outcomes
Communicate with local patients	Maintain national access and audit history
Continue operating during outages	Support population-level learning

The current systems would remain authoritative for the events they originate. A laboratory remains responsible for the result it produced. A physician practice remains responsible for its notes and diagnoses. A pharmacy remains responsible for the prescription it dispensed.

The national architecture would preserve those original assertions and create a resolved view across them.

Information Would Be Submitted as Care Occurs

New diagnoses, medications, procedures, test results, and significant observations would be transmitted to the national architecture as they were created or updated.

The architecture would not wait until another provider requested a patient's records and then contact every prior system in real time. It would maintain the longitudinal history continuously.

If a source system were temporarily unavailable, its previously submitted information would remain available. When service is restored, new or corrected information would be synchronized.

Relevant longitudinal information should appear within the healthcare system the provider already uses.

Information Within the Provider's Existing Workflow

A cardiologist might receive a cardiovascular summary containing prior diagnoses, medications, imaging, procedures, and trends in relevant test results. An orthopedic specialist might receive a different view of the same longitudinal history. Both could examine the complete record when necessary and follow links to the original evidence.

The national architecture would determine what information might be relevant to the context. The existing system would present it as part of the provider's normal workflow.

This avoids forcing clinicians to learn another portal, remember another password, or search another database.

AI services: national and commercial

Existing healthcare systems could use AI services provided by the national architecture or integrate certified commercial models.

Competition should continue in developing provider interfaces, workflow automation, clinical decision support, and specialized AI. Vendors would compete on how effectively they help providers use information—not on their ability to keep patient information confined within a proprietary system.

When a source corrects information, both the original and corrected versions would remain traceable. When two providers disagree, the national history would preserve both assertions and flag the conflict for resolution.

AI could identify inconsistencies, but only an authorized person or an established clinical rule could resolve them. The resolved history would always be linked back to the original evidence.

Resilience During Outages

Patient care must not be interrupted when the national architecture is temporarily unavailable.

Existing systems should retain a current, locally accessible summary for established patients, including medications, allergies, major diagnoses, implanted devices, and other information required for safe treatment. Emergency procedures would permit controlled access when identity or normal authorization could not be completed.

The national architecture would improve completeness without becoming a single point of operational failure.

What may diminish over time?

As adoption increases, several existing activities should become smaller or disappear:

- repeated patient-history questionnaires;
- manual record requests and faxing;
- proprietary point-to-point interfaces;
- scanning outside documents into local records;
- manual reconciliation of medication and condition lists;
- repeated entry of demographic and insurance information;
- separate retrieval of records from multiple patient portals;
- provider staff assembling routine authorization packets;
- payer staff rekeying and validating submitted information.

The transition for vendors

Healthcare information-system vendors have significant investments in their products, customers, and specialized knowledge. They should participate in the transition rather than be treated as obsolete.

Vendors would adapt their systems to submit standardized events, receive resolved information, invoke approved AI services, and present relevant evidence to users. They could continue differentiating through workflow, usability, automation, specialty functionality, and service.

The change is not that current systems cease to matter. It is that no single provider's system should be expected—or relied upon—to represent the patient's complete history.

The national architecture would not replace the systems that operate healthcare organizations. It would provide the shared information foundation those systems have never been able to create independently.

Reducing Administrative Waste and Fraud

The financial opportunity begins with the administrative exchange between providers and payers. The 2024 CAQH Index⁵ estimated that **administrative workflows tracked by the Index cost approximately \$90 billion annually, with about \$20 billion potentially avoidable through greater automation.** These workflows include activities such as eligibility verification, prior authorization, claims processing, and payments. These estimates address primarily how information is exchanged, not the additional effort required to find, interpret, validate, and resubmit information scattered across multiple systems.

Improper payments suggest another substantial opportunity, although they should not be confused with fraud. For fiscal year 2025, CMS estimated⁶ approximately \$95.5 billion in improper payments across Medicare fee-for-

⁵ CAQH. *2024 CAQH Index*. 2025.

https://www.caqh.org/hubfs/Index/2024%20Index%20Report/CAQH_IndexReport_2024_FINAL.pdf. The Index tracks approximately \$90 billion in annual spending on manual and automated administrative workflows in the medical and dental industries. It identifies a \$20 billion opportunity—about 22% of those costs—from shifting manual processes to automated workflows.

⁶ Centers for Medicare & Medicaid Services. “Fiscal Year 2025 Improper Payments Fact Sheet.” January 15, 2026. <https://www.cms.gov/newsroom/fact-sheets/fiscal-year-2025-improper-payments-fact-sheet>. CMS estimated FY2025 improper payments of \$28.83 billion for Medicare Fee-for-Service, \$23.67 billion for Medicare Advantage, \$4.23 billion for Medicare Part D, \$37.39 billion for Medicaid, and \$1.37 billion for CHIP—a combined \$95.49 billion. CMS emphasizes that improper-payment measurement is not a measure of fraud.

service, Medicare Advantage, Medicare Part D, Medicaid, and CHIP. These include documentation deficiencies, coding and eligibility errors, overpayments, underpayments, and payments that may involve fraud. GAO cautions⁷ that improper-payment estimates are not designed to measure fraud and cannot be used to estimate its extent.

The National Health Care Anti-Fraud Association⁸ cites estimates ranging from 3% to 10% of total healthcare spending, but no validated national estimate of fraud across government programs and private insurance exists. A resolved longitudinal health history would allow authorized systems to connect what was ordered, what was authorized, what service was documented, what result was produced, what was billed, and what was paid. That could reduce incomplete authorization requests, unnecessary denials, corrected claims, appeals, duplicate payments, billing unsupported by clinical records, and suspicious activity spread across multiple providers and payers.

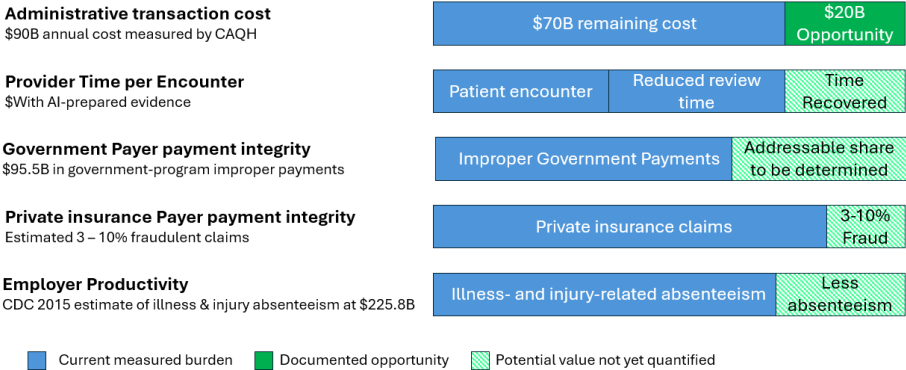
Better healthcare can produce value outside the healthcare system. Earlier diagnosis, more effective treatment, fewer complications, and faster recovery could reduce employee sick days, disability leave, and productivity loss. At the March 2026 average private-industry compensation cost, each avoided eight-hour absence represents approximately \$373 of productive capacity. The amount attributable to the proposed architecture cannot yet be estimated, but employer productivity and disability outcomes should be included when evaluating its potential benefits.

⁷ U.S. Government Accountability Office. *Improper Payments: Key Concepts and Information on Programs with High Rates or Lacking Estimates*. GAO-24-107482, June 27, 2024. <https://www.gao.gov/assets/gao-24-107482.pdf>. GAO explains that all fraudulent payments are improper, but not all improper payments result from fraud. Improper-payment estimates are not intended to measure the extent of fraud, and the reviews used to produce them are not designed to identify many forms of fraud.

⁸ National Health Care Anti-Fraud Association. “The Challenge of Health Care Fraud.” <https://www.nhcaa.org/tools-insights/about-health-care-fraud/the-challenge-of-health-care-fraud/>. NHCAA estimates that health care fraud losses amount to tens of billions of dollars annually, citing a conservative estimate of 3% of total health expenditures and estimates from some government and law-enforcement agencies as high as 10%. It also notes that fraudulent billings may be spread across multiple private insurers and public programs, making them less likely to be detected by any single payer.

Potential Measurable Value

Green identifies potential savings or productive time – not a guaranteed reduction



These estimates cannot simply be added together because the categories overlap, and not every improper payment represents a recoverable loss. They do, however, establish a potential savings range large enough to warrant serious evaluation.

Privacy Implications

A national longitudinal health history would become one of the most valuable—and most dangerous—collections of personal information ever created. Privacy and security cannot be added as features after the architecture is built. **They must determine how it is built, who can use it, and what happens when that trust is violated.**

Distributed Does Not Mean Private

Today, a person’s health information is distributed among physician practices, hospitals, laboratories, pharmacies, imaging centers, insurers, clearinghouses, cloud providers, and government programs.

That distribution may limit what some individual breaches expose. It also creates more copies, more interfaces, more credentials, and more people who can access portions of the information. Each organization may apply different security practices and maintain different audit records.

Most patients cannot determine who examined their information, what was viewed, why it was accessed, or whether it was subsequently copied elsewhere.

Fragmentation is not the same as privacy.

A consolidated architecture increases concentration risk, so the design should be logically integrated yet **physically segmented**. No single stolen credential, system administrator, or compromised server should expose an entire national database.

Access Based on Purpose

Access should not be granted simply because someone works for a provider, payer, or government agency. It should be limited by role, relationship, purpose, and time.

User	Permitted access
Individual	Complete history, corrections, disclosures, and access log
Treating provider	Information reasonably relevant to treatment
Specialist	Relevant history plus requested supporting detail
Emergency provider	Necessary emergency information through audited “break-glass” access
Medical-office staff	Minimum information needed for scheduling, referrals, authorization, or claims
Payer	Minimum clinical and administrative information necessary for coverage, authorization, and payment
Quality analysts	Deidentified or strongly protected information
Researchers	Approved data appropriate to the study, normally deidentified
System administrators	Technical access without routine ability to view identifiable clinical content
Government personnel	No access merely because the government regulates or funds the architecture

Every authorization should expire when the treatment, employment, or administrative relationship ends.

Every Access Should Be Visible

HIPAA already requires covered organizations to employ audit controls. Still, it does not provide patients with a simple, comprehensive history of every routine access for treatment, payment, and health care operations. The proposed architecture should go further.

Every retrieval, display, download, transmission, correction, and AI analysis involving identifiable information should create a tamper-resistant record containing:

- ✓ who accessed the information;
- ✓ the organization represented;
- ✓ the date and time;
- ✓ the information accessed;
- ✓ the declared purpose;
- ✓ the authority permitting access;
- ✓ whether information was downloaded or transmitted;
- ✓ whether AI processed the information;
- ✓ whether emergency access was invoked.

Individuals should be able to review this history as easily as they review credit card transactions. Unusual access should generate an alert.

If a nurse in Florida accesses my record while treating me, I should be able to see it. If someone in another state accesses it at two o'clock in the morning with no treatment, payment, or operational relationship to me, both the system and I should want an explanation.

Permitted and Prohibited Uses

Identifiable information should be used only for the individual's treatment, payment, and authorized healthcare operations. Broader analysis should use de-identified or strongly protected data to improve healthcare quality, public health, and approved research.

Information collected for healthcare must not be repurposed to disadvantage the individual.

Federal legislation should specifically prohibit use of the architecture for:

- ⊗ employment decisions;
- ⊗ credit decisions;

- ⊗ marketing without explicit authorization;
- ⊗ sale of identifiable information;
- ⊗ unrelated commercial profiling;
- ⊗ life or disability insurance underwriting without explicit authorization;
- ⊗ political targeting;
- ⊗ unauthorized surveillance;
- ⊗ unrestricted law-enforcement searches;
- ⊗ discrimination based on predicted future illness;
- ⊗ training commercial AI models outside approved purposes;
- ⊗ reidentifying protected analytical data.

Any legally compelled access should require narrowly defined authority, retrieve only the necessary information, and leave a permanent audit record. The document can recommend stricter judicial oversight without attempting to resolve every legal exception.

Congress should enact legislation specifically governing the national health information architecture. Unauthorized access, intentional misuse, sale, alteration, destruction, or reidentification of protected information should constitute a federal crime, whether committed by an outside hacker, healthcare employee, payer, contractor, or government official.

The legislation should include:

- § criminal penalties proportionate to intent and harm;
- § greater penalties for trusted insiders and organized attacks;
- § personal liability for knowingly misusing access;
- § mandatory prosecution referral standards;
- § civil penalties for negligent security;
- § compensation and remedies for affected individuals;
- § whistleblower protection for reporting misuse;
- § mandatory breach notification;
- § independent investigation and public reporting;
- § prohibition against contractual waivers of patient rights;
- § uniform national minimum protections while allowing stronger state protections;
- § penalties for organizations that conceal breaches or fail to review suspicious access.

Security Architecture

The technical safeguards should include:

- ◆ zero-trust authentication;
- ◆ multifactor authentication resistant to phishing;
- ◆ encryption in transit and at rest;
- ◆ separate encryption keys for segmented data;
- ◆ isolation of identifiers from clinical content where practical;
- ◆ purpose-based and time-limited access tokens;
- ◆ continuous detection of unusual access;
- ◆ independent penetration testing;
- ◆ immutable, separately protected audit records;
- ◆ rapid credential revocation;
- ◆ geographically separated recovery systems;
- ◆ local availability of essential clinical summaries during outages;
- ◆ no single administrator capable of accessing or exporting everything;
- ◆ multiple-person approval for extraordinary data operations.

AI systems should receive only the data required for an approved task.

Prompts, outputs, and model access should be logged. A model should not retain identifiable information for unrelated training.

The Privacy Trade

No architecture can promise that health information will never be attacked or misused. Today's distributed systems cannot make that promise either.

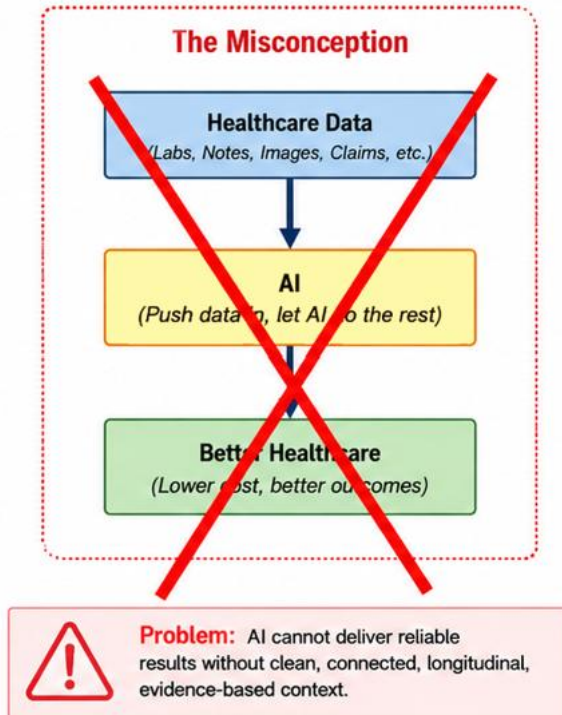
The appropriate question is whether a national architecture can provide stronger security, narrower access, more consistent enforcement, and far greater visibility than the fragmented environment that exists today.

Privacy should not mean that my information is scattered in a way that neither my physician nor I can find it. Privacy should mean that the right information is available for my care, unavailable for improper purposes, and accompanied by a permanent record of everyone who used it.

A national health information architecture would require the public to place substantial trust in it. That trust must be earned through technical design, enforceable restrictions, individual visibility, and criminal accountability—not requested through another eight-page form.

AI Alone Won't Transform Healthcare

It Requires an Information Architecture



The more I studied my proposal to use AI to streamline healthcare, the harder it became.

My proposal for a national longitudinal patient history didn't begin with technology. It began with observations.

Like many families, mine has repeatedly encountered redundant medical forms, fragmented information spread across multiple healthcare systems, delays caused by prior authorization, and specialists who often appeared to begin their investigations with little visibility into years of prior evaluations, unsuccessful treatments, and clinical history.

Those experiences led me to ask a simple question: Could a national longitudinal patient history, combined with AI, reduce administrative costs while helping physicians make better-informed decisions?

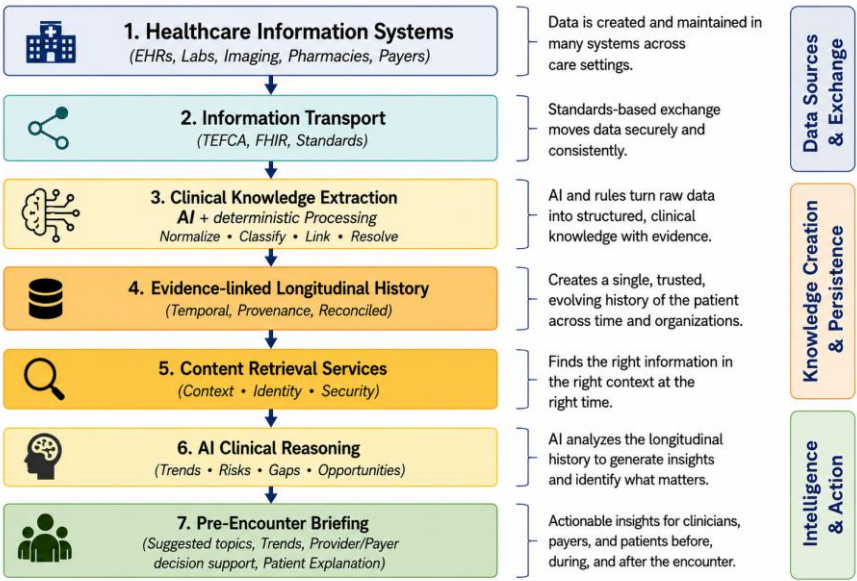
The proposal outlined that vision, described the potential benefits, and acknowledged obvious challenges such as privacy, security, governance, and patient consent.

What I didn't fully appreciate was that AI is only one component of the solution. Building a trustworthy longitudinal patient history requires an information architecture that collects, transports, reconciles, preserves, retrieves, and reasons over healthcare information before AI can add meaningful value.

Rather than weakening my conviction, that exercise strengthened my appreciation for the scale of the problem. It reinforced my belief that the idea deserves a structured feasibility study by experts from multiple disciplines.

The discussion that follows examines each layer of that architecture. Every layer introduces its own technical and organizational challenges, and each deserves careful study before any implementation effort begins.

The Vision: An Information Architecture for AI-Enabled Healthcare



1. Healthcare Information Systems

Healthcare information doesn't arrive in one convenient format. It is distributed across EHRs, laboratories, imaging systems, pharmacies, and payers. Some information is highly structured (medications, laboratory values, procedures), while other information exists only in free-form clinical notes, imaging reports, scanned documents, or medical images. Any national longitudinal history must integrate all of these forms of information.

2. Information Transport

TEFCA, FHIR, IHE profiles, and other interoperability standards provide many of the capabilities needed to exchange healthcare information across organizations. A continuously maintained longitudinal history would build on these capabilities. Still, it would require reliable notification or detection of new, corrected, and deleted information, along with mechanisms to detect missed updates and recover from communication failures.

3. Clinical Knowledge Extraction

Combining information from multiple organizations introduces ambiguity. One provider may record a diagnosis while another explicitly rules it out. Clinical notes may contradict structured diagnosis codes. The system must preserve provenance, supporting evidence, and temporal history while determining the current clinical interpretation rather than simply overwriting one answer with another.

4. Evidence-Linked Longitudinal History

A longitudinal history is more than an interoperable collection of records. It should preserve clinical assertions, supporting evidence, revisions, corrections, provenance, and the evolution of a patient's history over time. Technologies such as temporal databases may play an important role in preserving both clinical truth and historical understanding.

5. Context Retrieval

AI cannot reason effectively without context. The challenge is determining which portions of a patient's history are relevant to each new laboratory result, imaging study, physician note, or clinical question. Retrieving too little context risks missing important relationships, while retrieving everything would be inefficient and potentially overwhelming.

6. AI Clinical Reasoning

AI should not replace physician judgment. Instead, it should analyze the longitudinal history to identify significant trends, conflicting information, possible medication interactions, missing follow-up, and other issues that deserve clinical review. Every conclusion should remain linked to supporting evidence and explainable by both clinicians and patients.

7. Pre-Encounter Briefing

The objective is not another alert system. The goal is to prepare both physician and patient before the encounter by highlighting the few issues most deserving of discussion, reducing the time spent searching for information, and increasing the time available for clinical decision-making.

One conclusion has become increasingly clear to me: AI is not the architecture. AI is one component of a much larger information architecture. The real challenge is creating a trusted, evidence-linked, longitudinal patient history that provides AI with the context it needs to reason effectively.

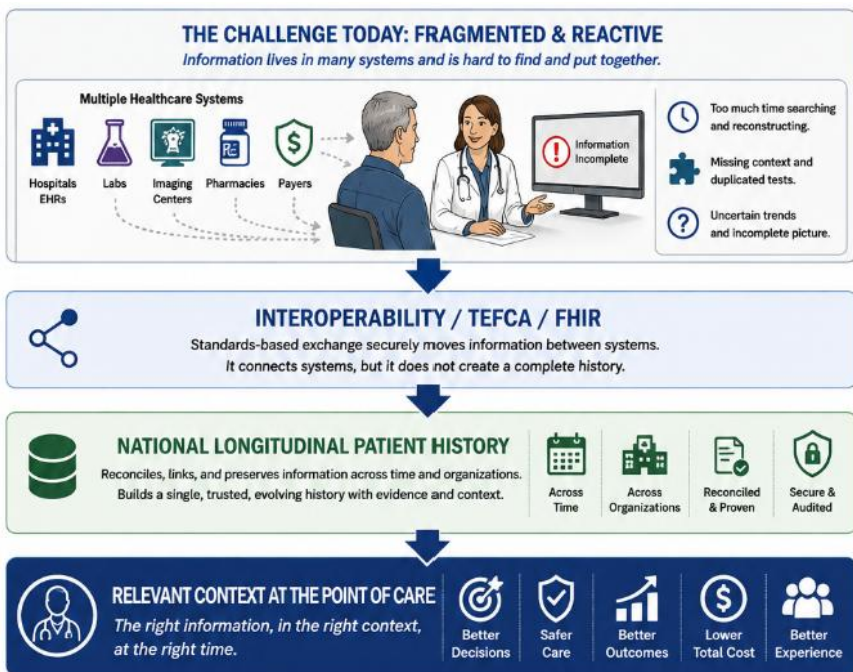
That realization hasn't weakened my confidence in the vision. Quite the opposite. It has convinced me that the opportunity is even greater than I originally imagined—and that the engineering challenges deserve the attention of experts from healthcare, information architecture, interoperability, artificial intelligence, privacy, security, and public policy.

That's exactly why my proposal recommended a focused feasibility study rather than immediate implementation.

Interoperability Is Not the Same as Longitudinal History

Why Healthcare AI Requires More Than Interoperability

This chapter examines why healthcare interoperability alone cannot create a trustworthy longitudinal patient history. TEFCA, FHIR, IHE profiles, and existing Healthcare Information Systems provide important capabilities for exchanging and accessing information across organizations. The missing architectural question is how that information can be assembled, reconciled, maintained, secured, and transformed into longitudinal knowledge available for better-informed decisions.



*Interoperability moves data.
Longitudinal History creates knowledge.*

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Introduction

The national longitudinal healthcare information architecture does not replace today's healthcare information systems. It enriches them by allowing each system to focus on what it does best while sharing capabilities that are better developed once and used by all.

The preceding chapters established the vision and the architectural premise behind it. A national longitudinal patient history, combined with AI, could reduce healthcare costs while supporting better-informed patient care. But AI is not the architecture. It is one component of a broader information architecture, and before AI can reason effectively, it needs a trustworthy, evidence-linked longitudinal patient history.

This chapter steps back one layer in that architecture to examine where that history comes from and how existing healthcare information systems and interoperability capabilities fit into the solution.

Health Care Information Systems

Modern healthcare information systems have transformed the way healthcare organizations operate. They support scheduling, registration, physician documentation, laboratory processing, imaging, pharmacy, billing, claims, and countless other operational processes that allow hospitals and physician practices to function efficiently.

At the same time, another transformation has occurred. The volume and diversity of clinical information about each patient has grown dramatically.

Today's healthcare generates an unprecedented volume of information, including:

- laboratory results
- imaging studies
- physician notes
- specialist consultations
- prescriptions
- wearable devices
- home monitoring
- genomics
- patient-generated information

Healthcare doesn't lack information. It lacks integrated, trustworthy patient history.

A typical patient may receive care from a primary care physician, multiple specialists, hospitals, imaging centers, laboratories, pharmacies, and other providers. As people age, relocate, or change insurance, their medical history becomes distributed across an increasing number of healthcare information systems. As a result, no single organization typically possesses a complete longitudinal view of the patient's health history.

The challenge is no longer simply collecting healthcare information. It is transforming information scattered across many organizations into a trustworthy longitudinal patient history, then helping physicians identify the small portion of that history that matters for today's clinical decision.

This is precisely the problem that existing healthcare information systems, interoperability capabilities, and a national longitudinal patient history should work together to solve. Each has a different role within the overall information architecture.

Building a Foundation for Healthcare Interoperability

Healthcare providers have long struggled to exchange patient information across organizational boundaries. Every hospital, physician practice, laboratory, imaging center, and pharmacy typically maintains its own healthcare information system. When patients receive care from multiple organizations, those systems must locate and exchange relevant information while protecting patient privacy.

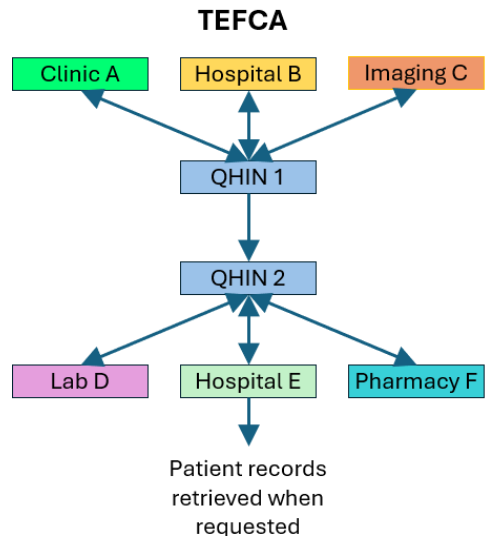
Nationwide healthcare information exchange did not begin with TEFCA. Federal initiatives such as the Nationwide Health Information Network (NHIN), together with standards and implementation profiles developed by organizations such as HL7 and Integrating the Healthcare Enterprise (IHE), have advanced the ability to locate, exchange, and securely access patient information across organizational boundaries.

The Trusted Exchange Framework and Common Agreement (TEFCA) builds on that progress by establishing common policies, technical requirements, and trusted networks through which participating organizations can discover and retrieve patient information from one another.

TEFCA is now operating at substantial scale⁹. Created in response to the 21st Century Cures Act, the framework was published in 2022 and became operational in December 2023. By 2026, more than 71,000 sites or organizations were participating through 11 QHINs, with 464 million documents exchanged by the end of 2025.

Healthcare organizations connect through **Qualified Health Information Networks (QHINs)**, which communicate with one another, allowing authorized providers to locate and retrieve patient information from participating organizations nationwide.

TEFCA represents a significant milestone in healthcare interoperability, allowing authorized providers to discover where patient information exists and retrieve it through a common nationwide framework rather than hundreds of proprietary interfaces.



But interoperability does not necessarily make the information usable.

Recent federal initiatives are expanding API-based access to healthcare information, including the CMS Health Technology Ecosystem. Commenting on that initiative, Dr. Mitesh Rao, CEO and cofounder of OMNY Health, observed that **“Interoperability doesn’t equal usability.”** Information obtained from different organizations can still be heterogeneous, incomplete, and difficult to use.¹⁰

Making that information usable requires a different architectural layer.

⁹ ASTP/ONC, **“Data Liquidity, Affordability, and Access: The History & Growth of TEFCA,”** updated June 5, 2026. <https://healthit.gov/resources/data-liquidity-affordability-and-access-the-history-growth-of-tefca/>

¹⁰ Healthcare IT News, **“CMS taps tech firms for new patient health data ecosystem.”** Reporting on the CMS Health Technology Ecosystem, Dr. Mitesh Rao, CEO and cofounder of OMNY Health, noted that “Interoperability doesn’t equal usability,” citing heterogeneity and incompleteness in healthcare data and the continuing need to transform exchanged data into a usable form.

Interoperability Is Not the Same as Longitudinal History

While TEFCA provides an essential interoperability framework, its purpose is fundamentally different from maintaining a continuously updated longitudinal patient history.

When a provider needs additional information about a patient, TEFCA can help locate and retrieve records held by participating organizations. Other interoperability standards and profiles provide additional exchange and notification capabilities. But interoperability does not, by itself, continuously reconcile information across organizations, preserve the clinical interpretation of conflicting evidence over time, or maintain a single trusted longitudinal patient history.

Those responsibilities belong to a different architectural layer.

A complete, trusted longitudinal patient history enables better decisions, safer care, and lower total cost.



The Missing Architectural Layer

A natural question is whether every healthcare information system could retrieve patient records through TEFCA and perform its own clinical knowledge extraction, reconciliation, longitudinal history management, and AI reasoning. Technically, it could.

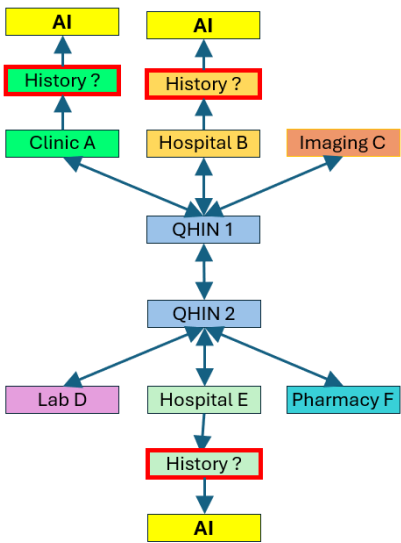
The architectural question is different: **Should every healthcare information system solve the same national problem independently?**

There is another important consideration. Today's interoperability framework provides mechanisms for exchanging and retrieving patient information, including some notification capabilities. But when new clinical information is created—or existing information is corrected—no nationwide mechanism

currently ensures that every healthcare information system maintaining its own longitudinal history receives the change. Without such a capability, each system would either have to repeatedly query TEFCA looking for changes or risk maintaining an incomplete or outdated longitudinal history.

A national longitudinal architecture changes that model. Rather than requiring hundreds of healthcare information systems to independently discover, retrieve, interpret, reconcile, and maintain the same patient history, participating systems would publish new and corrected clinical events through a robust change-data-capture interface. The longitudinal architecture would continuously transform those events into a trusted, evidence-linked patient history while preserving provenance, temporal history, and supporting clinical evidence.

Independent Longitudinal History



How are these histories kept synchronized?

That approach allows healthcare information systems to continue focusing on the functions they perform exceptionally well: scheduling, physician workflow, documentation, laboratory processing, imaging, billing, claims, and patient interaction. At the same time, they gain access to a continuously maintained longitudinal patient history and shared AI services that would otherwise need to be developed and maintained independently by every vendor.

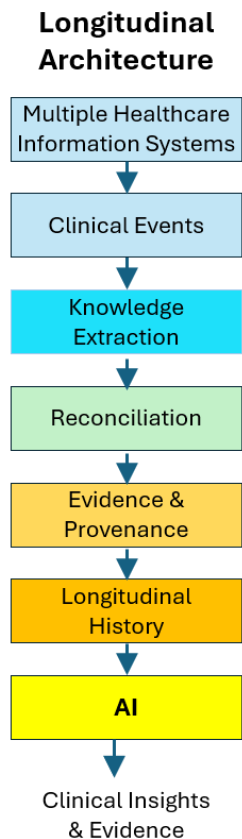
The National Longitudinal Healthcare Architecture

Rather than requiring every healthcare information system to independently discover changes, reconcile conflicting information, maintain its own longitudinal patient history, and develop AI capabilities, these responsibilities become shared services within a national information architecture.

As participating healthcare information systems create new clinical information or correct existing records, they publish those clinical events to the longitudinal architecture. Structured clinical information can be processed directly, while clinical notes, imaging reports, and other unstructured information undergo clinical knowledge extraction to identify diagnoses, observations, procedures, medications, and other clinically relevant facts together with the supporting evidence from which they were derived.

Information arriving from multiple organizations is then reconciled. Rather than overwriting conflicting information, the architecture preserves provenance, supporting evidence, temporal history, and differing clinical assertions so that the longitudinal history reflects both the patient's current status and the evolution of that history over time.

The resulting longitudinal patient history becomes the foundation for AI-assisted clinical reasoning. Because AI is now operating on an integrated, evidence-linked patient history rather than isolated records retrieved during



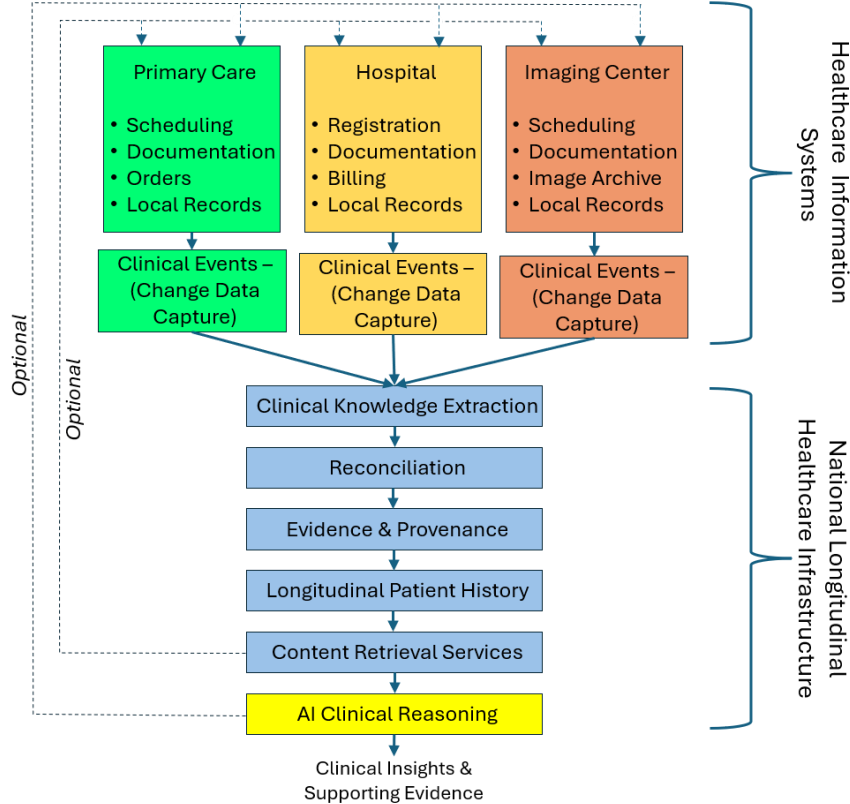
an encounter, it can identify trends, inconsistencies, medication interactions, unresolved issues, and other observations that may warrant discussion between the physician and patient. The architecture produces evidence-supported clinical insights rather than replacing clinical judgment.

Healthcare information systems remain the systems of record for the clinical information they create. The national longitudinal architecture does not replace those systems; it continuously assembles, reconciles, and enriches their information into a trusted patient history that every participating healthcare organization can share.

Impact on Healthcare Information Systems

A national longitudinal healthcare architecture is intended to complement—not replace—today's healthcare information systems. Participating systems remain responsible for the operational functions that support hospitals, physician practices, laboratories, imaging centers, pharmacies, and payers. The national infrastructure assumes responsibility only for capabilities inherently shared across organizations: assembling, reconciling, and maintaining a trustworthy longitudinal patient history while making that history available through common retrieval and AI services.

Division of Responsibility Within a National Healthcare Architecture



Participating healthcare information systems would continue to create and maintain the clinical records required to operate physician practices, hospitals, laboratories, imaging centers, pharmacies, and payer organizations. They remain the authoritative systems of record for the clinical information they generate.

The national longitudinal healthcare infrastructure maintains the integrated, evidence-linked longitudinal patient history, while participating healthcare information systems maintain the operational records required to deliver care. This clear division of responsibility allows each layer of the architecture to focus on what it does best.

To support a continuously maintained longitudinal patient history, participating systems would provide a robust Change Data Capture interface that makes newly created, corrected, and deleted clinical information available to the national longitudinal healthcare infrastructure. The proposed

interfaces would build upon existing interoperability capabilities—including TEFCA, FHIR, IHE profiles, and related standards—rather than replacing them. The primary addition is a reliable mechanism for communicating changes as clinical information is created or revised, keeping the longitudinal patient history continuously synchronized with participating healthcare information systems.

Supporting continuous synchronization requires more than simply transmitting clinical data. Clinical events must include persistent identifiers so that subsequent corrections or revisions can be associated with the original event rather than treated as entirely new information. Similar persistent identifiers may also be required for evolving collections of clinical information such as allergies, medication lists, patient permissions, implanted devices, and other clinical assertions whose contents change over time.

Once the national infrastructure has assembled and maintained the longitudinal patient history, participating healthcare information systems have flexibility in how they use it. Some organizations may retrieve relevant portions of the patient's history using **Content Retrieval Services**. Others may also leverage nationally provided AI Clinical Reasoning services to analyze and summarize the retrieved knowledge, identifying trends, conflicting information, medication interactions, unresolved follow-up, and other topics deserving further clinical review. Organizations that want to differentiate their products can still develop their own AI capabilities, using longitudinal patient history as a trusted foundation.

Perhaps the greatest benefit to healthcare information system vendors is that they no longer need to independently develop and maintain many of the complex capabilities associated with longitudinal history management. Clinical knowledge extraction, reconciliation of conflicting information, provenance management, temporal history, maintenance of a trusted longitudinal patient history, and shared AI services become national infrastructure rather than capabilities every vendor repeatedly implements. Healthcare information systems can then focus on the areas where they provide the greatest value: intuitive user interfaces, efficient clinical workflows, scheduling, documentation, billing, practice management, and innovations that directly support healthcare providers and patients.

What This Could Look Like

Imagine arriving for your annual physical ten years from now.

Instead of completing several pages of forms listing surgeries, medications, specialists, allergies, and family history that have been entered countless times before, your physician's healthcare information system has already retrieved a trustworthy longitudinal patient history assembled from every participating healthcare organization that has cared for you.

Since the patient's last encounter, new laboratory results, imaging studies, specialist consultations, medication changes, and hospital visits have been incorporated into the longitudinal history as they became available. Conflicting information has been reconciled while preserving the original evidence and its source. Significant trends have been identified. Questions that deserve discussion have already been highlighted.

Before you enter the examination room, your physician receives a concise briefing rather than hundreds of pages of records.

"The patient's LDL cholesterol has increased steadily over the past four years despite medication."

"Blood pressure has remained well controlled."

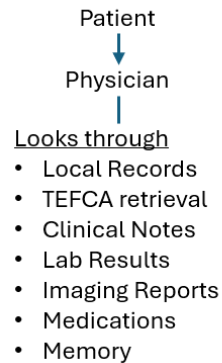
"A pulmonary nodule first identified three years ago has remained stable across four imaging studies."

"A medication prescribed by one specialist may be contributing to symptoms reported to another."

"Family history suggests additional screening may be appropriate."

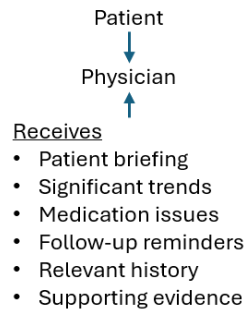
Rather than spending valuable appointment time searching for information or reconstructing years of medical history, physicians and patients begin the visit discussing the issues that matter most.

Today



Appointment time spent searching

Future



Appointment time spent making decisions

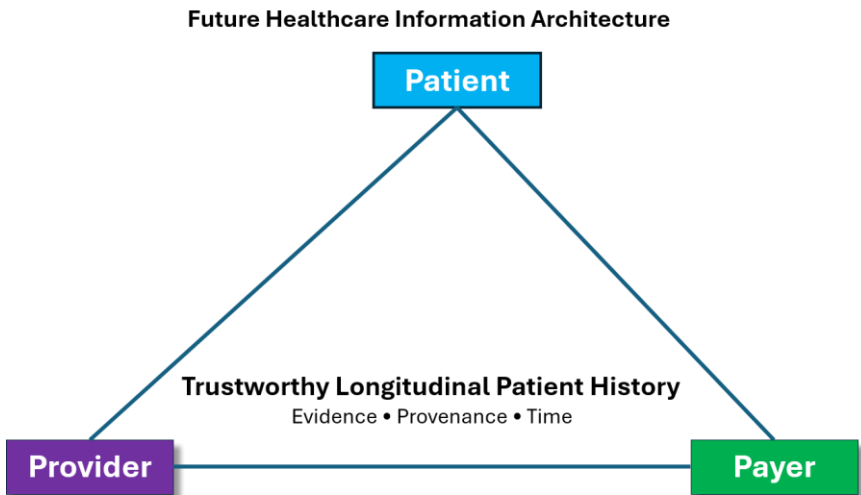
That is the opportunity.

The objective is not to replace physicians, healthcare information systems, or TEFCA. It is to build upon them by creating a national information architecture that transforms fragmented records into a trusted longitudinal patient history—allowing healthcare professionals to spend less time finding information and more time caring for patients.

Building a Trustworthy Longitudinal Patient History

From Healthcare Data to Clinical Knowledge

A national longitudinal patient history requires more than exchanging healthcare records. This chapter describes the information needed to create that history and how structured data and clinical documents can be transformed into evidence-linked, longitudinal clinical knowledge while preserving conflicting information, provenance, and temporal context.



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Introduction

The previous chapter described an important limitation of today's healthcare interoperability architecture. TEFCA, FHIR, IHE profiles, and related interoperability capabilities support exchanging and retrieving information across healthcare organizations, including some notification capabilities. But a longitudinal patient history must remain current as new clinical events occur, and existing information is corrected or changed. This requires a reliable mechanism for participating systems to communicate those changes to the longitudinal infrastructure and ensure they are received, rather than reconstructing the information when needed.

If that information can be continuously assembled, the next question is what the resulting longitudinal patient history would contain and how it could improve healthcare decisions and reduce costs.

Technical teams designing information systems commonly use **logical data models** and **physical data models** to define information in considerable detail, including individual data elements, relationships, keys, and how the data will ultimately be stored. That level of detail is not necessary to understand the proposed architecture.

Instead, this chapter uses a **conceptual information model**. It identifies the major types of information available in a longitudinal patient history and the important relationships among them. For example, a patient may have multiple medical conditions; those conditions may be assessed during multiple clinical encounters; an encounter may produce clinical notes, observations, laboratory results, imaging studies, procedures, or diagnoses; and procedures may subsequently result in claims and payments.

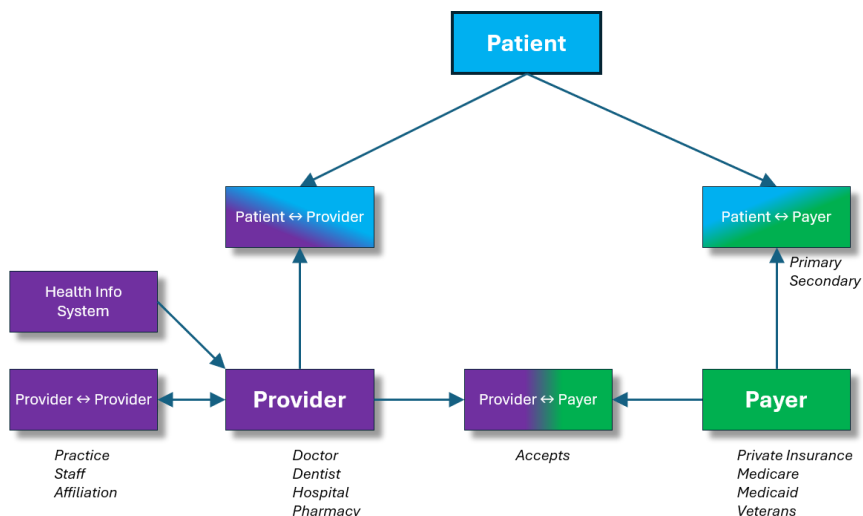
Understanding these relationships helps explain what becomes possible when information that is currently fragmented across healthcare organizations is assembled into a longitudinal history. It also helps distinguish information that can be obtained directly from existing healthcare systems from clinical knowledge that must be extracted, reconciled, and maintained over time.

The following diagrams progressively build the conceptual model, beginning with the patient, provider, and payer relationships and then adding the patient, clinical, and administrative information needed to create a useful longitudinal history.

The Healthcare Participants

At the highest level, healthcare involves three principal participants:

patients, providers, and payers.



A patient may receive care from many providers over time, while each provider serves many patients. The **Patient↔Provider** relationship represents this many-to-many relationship and provides a way to associate a patient with the providers and healthcare organizations involved in their care.

Similarly, a patient may be covered by different payers over time, and sometimes by more than one payer at the same time. The **Patient↔Payer** represents this relationship and allows coverage to be associated with the periods during which it applies.

Providers also have relationships with other providers and healthcare organizations. A physician may belong to a medical practice, a practice may be affiliated with a hospital, and hospitals, laboratories, imaging centers, and pharmacies may participate in care delivery. Provider ↔ Provider represents these relationships.

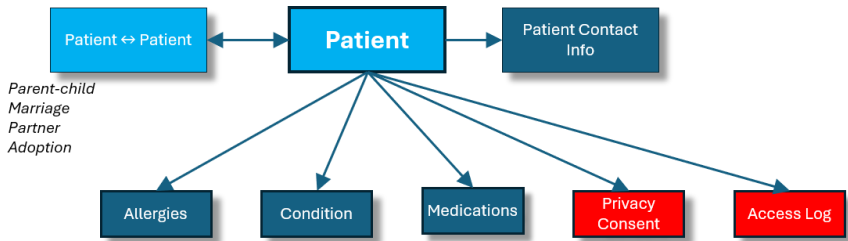
Finally, **Provider↔Payer** represents the relationships between providers and the payers that accept their insurance plans.

Existing Healthcare Information Systems remain associated with the providers that use them. They continue to support the operational functions

of those organizations while supplying information needed to build and maintain the longitudinal patient history.

The Patient

The longitudinal history is organized around the patient rather than around any individual healthcare organization.



Basic demographic and contact information identifies the patient, but several other types of information must follow the patient throughout their lifetime, regardless of where care is received.

Patient ↔ Patient represents relationships with other patients, such as parent-child, marriage or partnership, and adoption. These relationships may provide clinically relevant family context and support situations in which one individual is authorized to act on behalf of another.

Allergies identify substances that may cause an adverse reaction and need to be available whenever treatment decisions are made.

Conditions represent both current and historical health conditions. Some may be established through clinical care, while others may initially be known only because the patient reported them as part of their medical history.

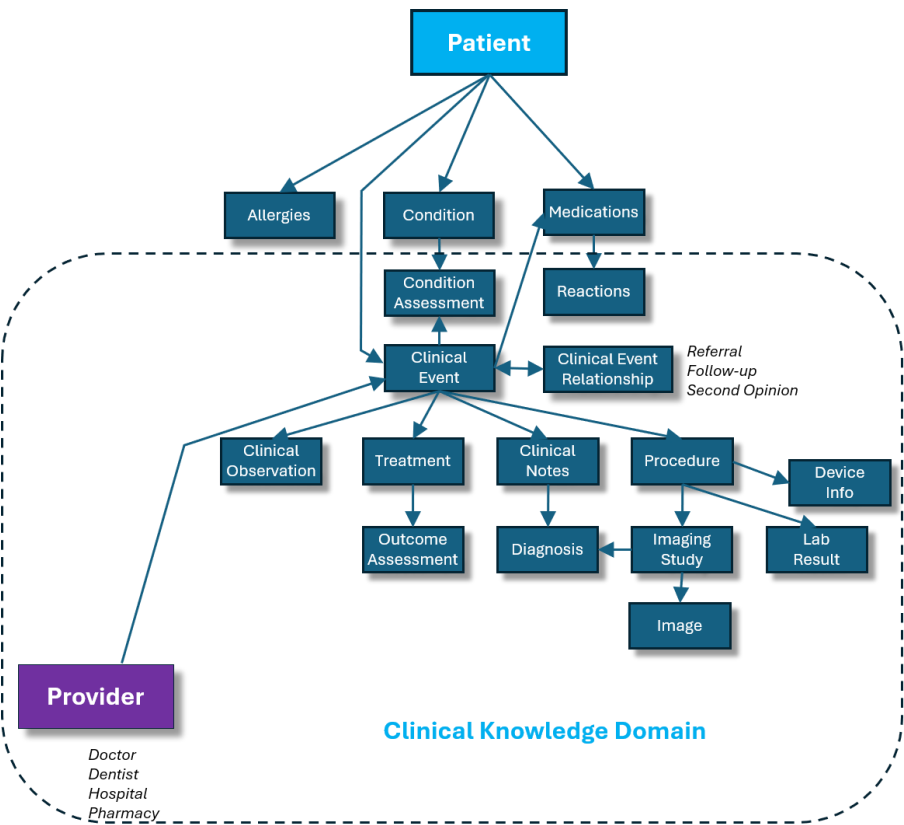
Medications provide a longitudinal record of medications the patient is taking or has taken. This includes medications prescribed by providers as well as over-the-counter medications reported by the patient. Maintaining medication information at the patient level allows it to persist independently of the individual clinical events in which medications may be prescribed, changed, discontinued, or reviewed.

Privacy Consent records the patient's choices governing access to their healthcare information. At the same time, the **Access Log** provides a history of who accessed that information, when it was accessed, and for what purpose.

Together, these elements establish information that belongs to the patient's longitudinal history rather than to any single provider, payer, or clinical encounter.

The Clinical Knowledge Domain

Most healthcare information is created during interactions between patients and providers. In this conceptual model, each encounter is represented as a **Clinical Event**. A routine physical, an emergency room visit, a specialist consultation, a hospital admission, labs, or a telehealth appointment are all examples of clinical events.



A clinical event may exist without being associated with a particular medical condition. A routine physical is one example. More commonly, however, one or more of the patient's conditions are evaluated during an encounter. Because a patient may have many conditions addressed during one clinical event, and a condition may be evaluated during many clinical events over

time, **Condition Assessment** represents the relationship between conditions and the clinical events in which they are evaluated.

A **Clinical Event** may also initiate or modify a **Treatment**. Treatments need to be connected to outcomes. An **Outcome Assessment** records whether the intended improvement occurred and identifies complications, side effects, recurrence, or other consequences. An outcome may be documented during a subsequent clinical event or reported by the patient. When no follow-up information is available, the outcome should remain explicitly unknown rather than assuming that absence of further treatment indicates success.

Clinical events produce much of the information traditionally maintained by Healthcare Information Systems. This may include **Clinical Notes, Clinical Observations, Diagnoses, Procedures, Laboratory Results, Imaging Studies, Images, and information about implanted Devices**. Providers may also prescribe, change, discontinue, or review medications during an encounter.

Not all of this information is equally structured. Laboratory results, medication orders, procedure codes, and many clinical observations may already be represented using defined fields and healthcare standards. Clinical notes and imaging reports contain additional knowledge expressed as narrative text. Clinical Knowledge Extraction must identify clinically significant information within those documents while retaining the source material and its provenance.

Conditions and Condition Assessments add longitudinal context to this source information. A diagnosis recorded during one encounter may identify a new condition, confirm an existing condition, modify an earlier assessment, or conflict with information recorded elsewhere. Rather than treating every diagnosis or mention in a clinical note as an independent fact, the longitudinal architecture must reconcile this information with what is already known about the patient.

For example, a condition might first appear as a patient-reported medical history, later be evaluated by a primary-care physician, and subsequently be confirmed, questioned, or refined by a specialist. The longitudinal history should preserve each assessment, when it occurred, who made it, and the evidence supporting it, rather than replacing the history with only the most recent conclusion.

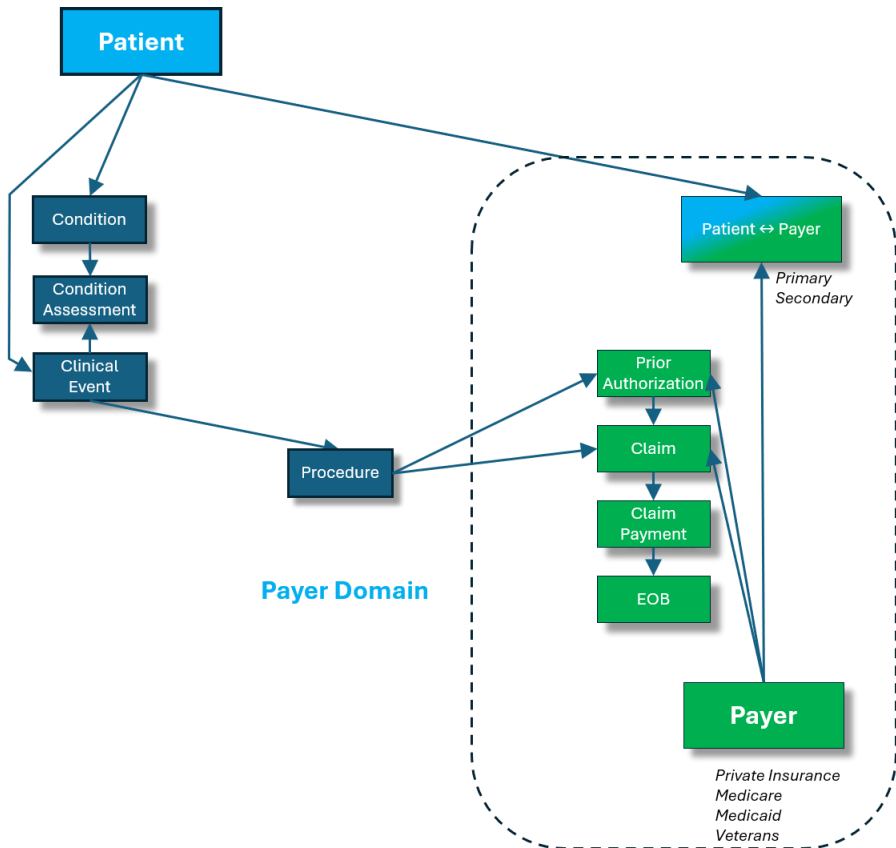
Clinical events may also be related to other clinical events. A primary-care visit may result in a referral to a specialist; a specialist consultation may lead to a procedure; and a procedure may result in subsequent follow-up visits.

Clinical Event Relationship preserves these connections so that related encounters can be understood as part of the patient's continuing care rather than as isolated events.

Together, these relationships transform a collection of healthcare records into a longitudinal clinical history. The original clinical information remains available, while extracted and reconciled knowledge helps providers understand what conditions the patient has had, how those conditions have changed, what treatments have been provided, what outcomes followed, and what evidence supports those conclusions.

The Payer Domain

Healthcare also produces information describing how care is authorized, billed, and paid. The **Payer Domain** connects this administrative information to the same patient and clinical history used by providers.



Patient ↔ Payer identifies the organizations responsible for a patient's healthcare coverage. Because coverage changes over time and patients may have both primary and secondary coverage, these relationships must retain the periods during which each coverage relationship was valid.

Some procedures require **Prior Authorization** before they can be performed. After services are provided, **Claims** identify the procedures for which payment is requested. **Claim Payments** record how those claims were adjudicated and paid, while the **Explanation of Benefits (EOB)** communicates the claim's disposition and the patient's financial responsibility.

Connecting this information to the clinical history provides capabilities that neither the clinical nor payer information can provide independently. Prior authorization can be supported by relevant clinical history rather than repeatedly assembling documentation from individual providers. Claims can be compared with the procedures and clinical information from which they originated, potentially reducing missing information, duplicate submissions, and manual review.

Over time, connecting clinical information with claims and payment information also makes it possible to examine the relationship between **care, cost, and outcomes**. Providers can better understand prior treatment and its results, patients can see a more complete history of the care they received, and payers can evaluate the cost and effectiveness of alternative treatments across comparable patient populations.

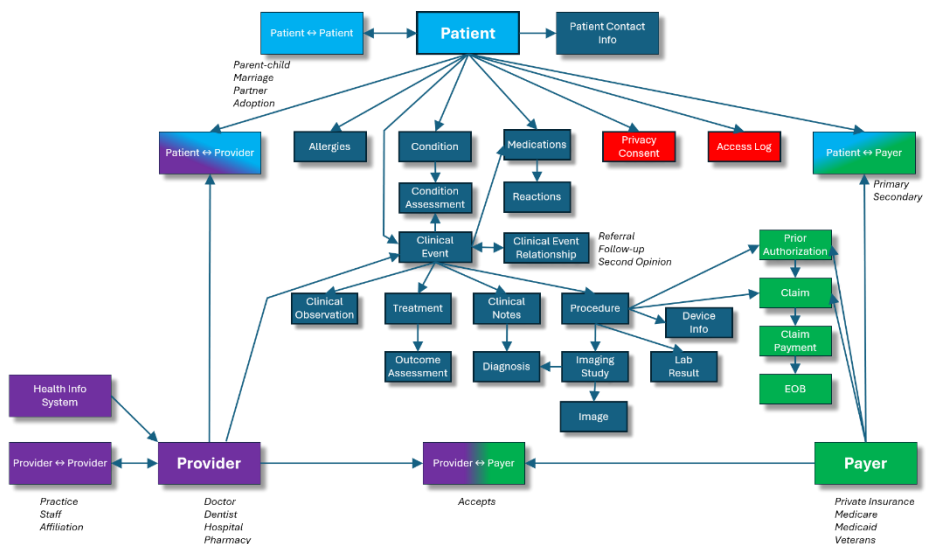
The objective is not to merge provider and payer operational systems. Each continues performing the functions for which it was designed. The longitudinal architecture provides the common patient context that allows clinical and administrative information distributed across those systems to be understood together.

Bringing the Information Together

The previous diagrams examined portions of the conceptual model separately so that their relationships could be seen more clearly. Together, they form a single longitudinal view of the patient.

The complete model connects the patient's conditions and clinical history with the providers who delivered care and the payers that authorized and paid for it. Information that today resides in separate Healthcare Information Systems and payer systems can therefore be related through the patient and the clinical events that occurred over time.

The value of the architecture does not come simply from collecting more data. It comes from preserving these relationships so that clinical information can be interpreted in context and used by providers, patients, and payers.



How the Longitudinal History Is Populated

The conceptual model describes the information available in the longitudinal patient history. Populating and maintaining that history requires continuously collecting information from Healthcare Information Systems and payer systems and transforming it into a consistent longitudinal representation.

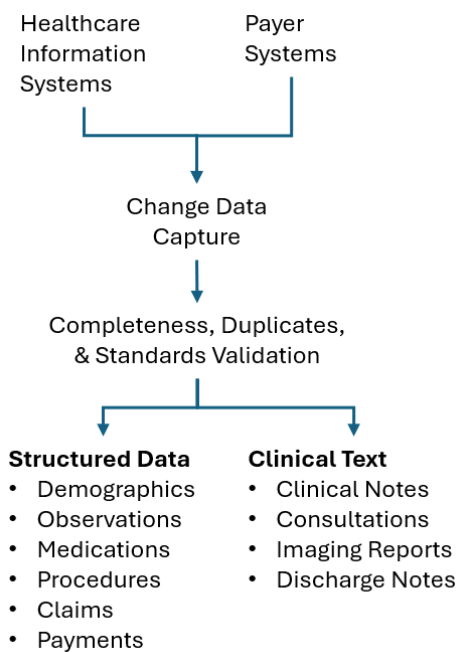
Not all this processing requires AI. Much of it uses well-established data integration techniques. AI becomes important when clinically significant knowledge must be extracted from narrative information or when information from multiple sources must be reconciled.

Receiving Complete and Reliable Data

The first step is conventional **Change Data Capture (CDC)**. Participating systems must identify information added, changed, or deleted since the previous exchange.

The receiving architecture must determine whether expected data was received, detect duplicate transmissions, recognize corrections to previously received information, and prevent retransmitted information from being interpreted as new clinical activity. Persistent source identifiers are important so that a corrected clinical note, laboratory result, claim, or other record can be associated with the information it replaces or modifies.

This processing is mundane compared with AI, but it is essential. No subsequent analysis can produce a trustworthy longitudinal history if source information is missing, duplicated, or incorrectly interpreted as new.



Validating and Standardizing Structured Information

Much of the information received from healthcare and payer systems is already structured. Patient demographics, clinical observations, medications, procedures, laboratory results, claims, and claim payments contain defined data elements that can be validated and normalized upon receipt.

Healthcare interoperability standards substantially simplify this task by defining formats, identifiers, terminology, and coding systems. The longitudinal architecture must nevertheless validate incoming values, recognize unsupported or locally defined values, normalize differences where appropriate, and retain the source representation.

Blood pressure, medication, procedure, or claim should not acquire a different meaning simply because it originated in a different Healthcare Information System.

Extracting Knowledge from Clinical Text

Some of the most important information about a patient is contained in narrative documents rather than structured columns. Clinical notes, discharge summaries, operative reports, specialist consultations, and imaging reports may describe conditions, symptoms, diagnoses, clinical observations, treatment decisions, treatment response, adverse effects, uncertainty, and historical information.

Large Language Models and other AI techniques can identify this information and transform clinically significant statements into structured knowledge.

The extracted information should not replace the original document. The complete Clinical Note remains part of the longitudinal history. Extracted knowledge retains a link to the source document—and, where practical, to the portion of the document supporting the extraction—so that a provider can verify how a conclusion was reached.

For example, a note might state that a patient's knee pain improved for several weeks following an injection but subsequently returned. Knowledge extraction could identify the condition under discussion, the procedure, the reported improvement, its duration, and any subsequent recurrence, while preserving the original statement as supporting evidence.

Identifying Redundant Information

Healthcare information is frequently repeated. A diagnosis may appear in a structured field, in a clinical note, in a specialist's report, and again in subsequent notes that repeat information from earlier encounters.

The longitudinal architecture should not interpret every occurrence as independent evidence.

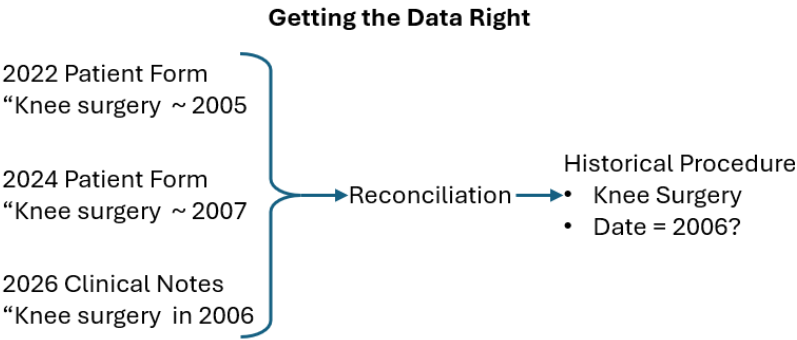
Patient identity, provider, clinical event dates, source identifiers, terminology, and other contextual information can be used to determine whether apparently similar information represents a new clinical observation, a confirmation from an independent source, or a repetition of information already received.

The source records remain available. The objective is not to delete duplication, but to prevent duplicated information from distorting the longitudinal knowledge derived from those records.

Reconciling Clinical History Over Time

A particularly important problem is determining how newly received information relates to the patient's existing Conditions and Condition Assessments.

A newly extracted diagnosis might identify a new condition. It might confirm or refine an existing condition. It might represent a historical condition reported by the patient. It might contradict an earlier assessment. It might also be uncertain or subsequently ruled out.



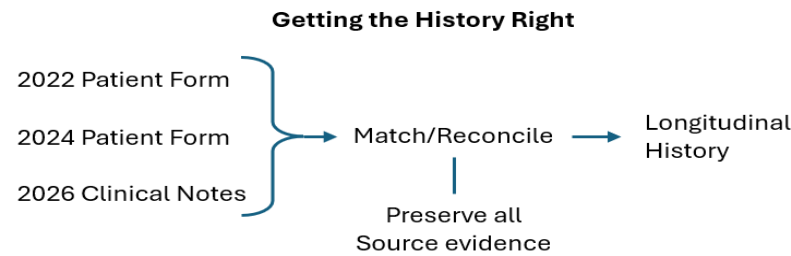
Patient-reported history presents a different reconciliation problem. Patients are frequently asked to provide previous medical conditions, surgeries, and approximate dates. Their recollection may change each time the information is requested. A surgery reported as occurring in 2005 on one questionnaire might be reported as occurring in 2007 several years later. These should not automatically be interpreted as two separate surgeries, nor should the later answer replace the earlier one.

The longitudinal history should preserve what was reported, when it was reported, and the source of the information while determining whether the

reports refer to the same historical event. If more authoritative evidence later becomes available—for example, the original surgical record—it can further refine the history without eliminating the earlier patient-reported information.

Preserving Time and Provenance

Healthcare information cannot be understood without knowing **when it was true and where it came from**.



The date the record was received is not necessarily the date of the clinical event. The date a provider documented a condition may differ from when the condition began. A clinical note written today may describe surgery performed twenty years ago. The medication may have been prescribed on one date, started later, changed several times, and eventually discontinued.

The longitudinal architecture therefore must preserve temporal context rather than reducing the patient's history to only the latest values.

It must also preserve provenance. Clinical knowledge should remain traceable to the provider, clinical event, source system, source document, and supporting evidence from which it was obtained.

Together, temporal context and provenance allow the longitudinal history to answer not merely **what is known about the patient**, but also **when it was known, who reported or assessed it, how that understanding changed, and what evidence supports it**.

From Data Collection to Clinical Knowledge

The result is more than a repository containing copies of healthcare records. Structured information is validated and normalized. Narrative information is converted into evidence-linked clinical knowledge. Repeated information is recognized. Conflicting assessments are preserved and reconciled. Changes

are maintained over time, and every significant conclusion remains traceable to its source.

That transformation is what allows information distributed across many Healthcare Information Systems and payer systems to become a usable longitudinal patient history.

Conclusion

A longitudinal patient history is more than a collection of records retrieved from different healthcare organizations. Its value comes from connecting information about patients, providers, clinical events, conditions, treatments, and payer activity while preserving when information was known, where it came from, and the evidence supporting it.

Much of this information already exists in Healthcare Information Systems and payer systems. Structured information can be validated and normalized, while AI can extract additional clinical knowledge from narrative documents. That information must then be matched, reconciled, and maintained over time without losing conflicting information or its provenance.

As discussed in the previous chapter, existing Healthcare Information Systems and interoperability capabilities—including TEFCA, FHIR, and IHE profiles—provide mechanisms for exchanging, retrieving, and in some cases notifying systems of changes in healthcare information. But maintaining a current national longitudinal history requires a reliable mechanism to ensure that relevant new, changed, and deleted information is communicated to the longitudinal infrastructure as it occurs.

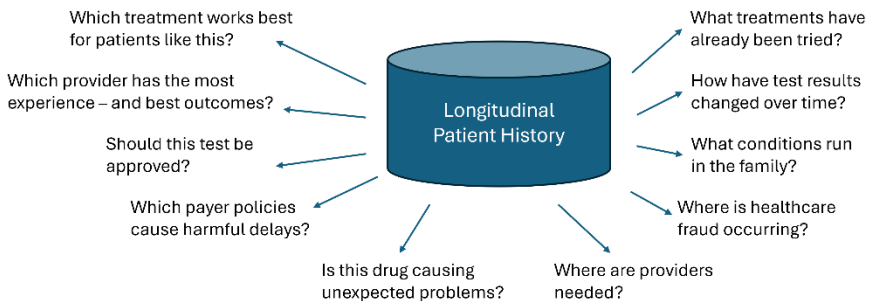
Creating a trustworthy longitudinal patient history is only part of the opportunity. Its value comes from how it can be used. The next chapter examines how that shared history could improve decisions and reduce administrative effort for patients, providers, and payers.

One History, Many Uses

REDUCING COSTS AND IMPROVING HEALTHCARE EFFECTIVENESS

A national longitudinal patient history can do far more than give providers better access to medical records. This chapter explores how the same trusted information can support patients, providers, payers, researchers, manufacturers, and government—using secure, purpose-based access and AI to retrieve and analyze the information appropriate to each need. These capabilities could improve clinical decisions and patient outcomes, identify more effective treatments and providers, streamline prior authorization, expose ineffective or costly practices, accelerate research, and reduce healthcare costs.

What Could We Learn From a Lifetime of Healthcare Data?



One History. Many Uses.

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Introduction:

The prior chapter described how information distributed across healthcare organizations could be assembled into a trustworthy longitudinal patient history. That history connects the patient with providers, clinical events, physician notes, diagnoses and conditions, laboratory results, medical images, medications, treatments, outcomes, and payer information while preserving the source evidence.

Why?

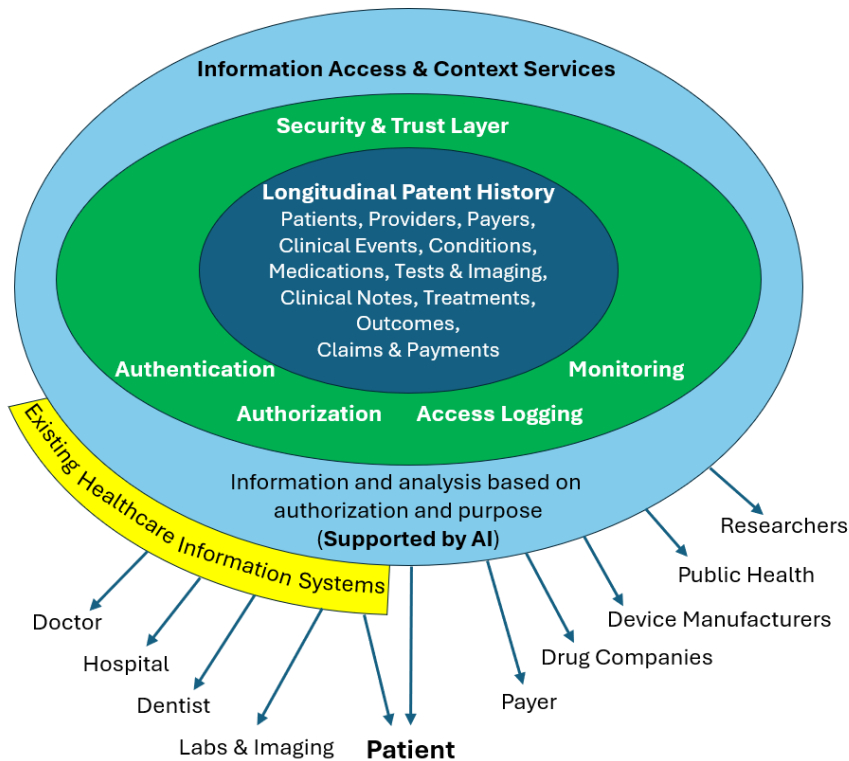
Imagine a healthcare information resource that could help:

- **Physicians** immediately find the patient history relevant to the decision in front of them.
- **Patients** find providers with the most experience—and best outcomes—treating patients like them.
- **Families** identify patterns of disease across generations.
- **Hospitals and providers** learn which treatments produce the best outcomes for comparable patients.
- **Researchers** study disease progression, treatments, and outcomes across millions of longitudinal histories.
- **Drug companies** identify differences in effectiveness and adverse effects across patient populations.
- **Device manufacturers** track the long-term performance of implants and medical devices.
- **Payers** accelerate appropriate care, reduce unnecessary administrative effort, and learn when authorization policies increase rather than reduce total healthcare costs.
- **Patients and regulators** compare payer authorization and denial practices—and their effects on cost and patient outcomes.
- **Healthcare policymakers** understand where trillions of healthcare dollars are producing value—and where they are not.

These are not separate information problems requiring separate national databases. They are different uses of the same longitudinal healthcare information.

With U.S. healthcare spending measured in trillions of dollars annually, even small improvements in administrative efficiency, unnecessary care, treatment effectiveness, adverse events, and provider selection can represent billions of dollars. The larger opportunity comes from addressing many of these opportunities simultaneously.

Realizing these benefits requires more than creating a longitudinal patient history. The architecture must make that information available for many different purposes while controlling who can access it, what they can access, and how it can be used.



One History Does Not Mean One Healthcare System

A national longitudinal patient history does not require a national healthcare delivery system, a national EHR, or federal control over healthcare decisions.

Hospitals, physician practices, laboratories, pharmacies, dentists, rehabilitation facilities, payers, and other healthcare organizations would continue to operate their own Healthcare Information Systems and make their own clinical and business decisions.

The proposed architecture instead provides information capabilities that are difficult for individual healthcare organizations to provide independently. It creates a longitudinal history that follows the patient across organizations and over time, while existing systems continue to support appointments, clinical documentation, orders, prescriptions, laboratory and imaging workflows, billing, claims, and the many other functions required to deliver and pay for healthcare.

This separation is important. No individual provider, payer, or technology vendor has a complete view of a patient's lifetime healthcare experience. Requiring every organization to independently locate, retrieve, reconcile, and interpret information from every other organization perpetuates duplication while still producing incomplete histories. A shared national information architecture performs these common functions once and makes the resulting information available to authorized users when needed.

The same principle applies to the broader uses described in this chapter. Researchers, pharmaceutical companies, device manufacturers, payers, and policymakers do not need their own copies of a national patient database. They need controlled services that can answer authorized questions using appropriate portions of the longitudinal information.

The proposal centralizes information capabilities, not healthcare decisions. Existing healthcare organizations continue to deliver and manage healthcare. The national architecture provides a common information foundation that allows them—and other authorized users—to make better-informed decisions.

One History Does Not Mean Everyone Sees Everything

Bringing longitudinal healthcare information together does not mean making it universally accessible. The architecture must determine who is requesting information, why they are requesting it, and what information they are authorized to receive.

Patients would have access to their own identified longitudinal history. Healthcare providers directly involved in a patient's care could receive identified information appropriate to that care. Other users—such as researchers, pharmaceutical companies, device manufacturers, and healthcare analysts—could be authorized to use selected clinical and demographic information without receiving information that identifies individual patients.

In many cases, these users would not need access to individual records. A researcher might need to identify patterns among patients with particular characteristics. A pharmaceutical company might examine treatment effectiveness or adverse events across patient populations. An implant manufacturer might analyze device longevity and revision rates. The architecture could perform these analyses on longitudinal history and return authorized summary results without providing copies of the underlying patient histories.

Security therefore involves more than protecting a database. Every request must be authenticated to establish who is making it, authorized to determine what that user is permitted to access, and constrained to information appropriate for the authorized purpose. Access should be logged so that there is an auditable record of who requested information, when it was requested, for what purpose, and what information was provided.

This creates an important distinction: **access to the national information resource does not necessarily mean access to patient records.** Different users can benefit from the same longitudinal information while receiving very different levels of detail.

One history can support many uses without giving every user the same view of that history.

Different Questions Require Different Information

The value of a longitudinal patient history does not come from giving every authorized user access to everything it contains. It comes from identifying the information needed for a particular purpose and using that information to answer a question or support a decision.

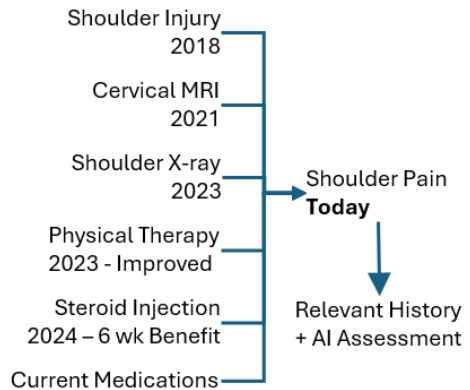
The Information Access & Context Services shown in the architecture provide this capability. After the Security & Trust Layer authenticates the user

and determines what information that user is authorized to use, the services identify the information relevant to the request. AI can help understand the request, identify relevant relationships within the longitudinal history, analyze the authorized information, and present useful results. AI does not determine what information the user is authorized to access.

Consider several very different requests:

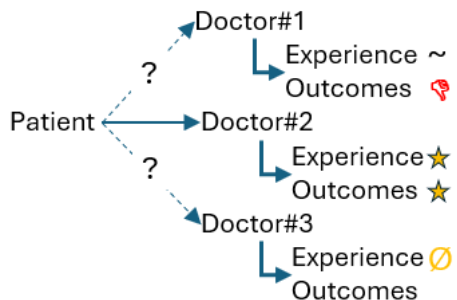
A physician evaluating a patient with recurring shoulder pain

might need prior shoulder and neck conditions, injuries, imaging, treatments, medications, physical therapy, procedures, treatment outcomes, and recent clinical observations. AI could assemble the relevant history across years of care, identify changes and prior unsuccessful treatments, and summarize the findings with links to the supporting clinical evidence.



A patient recently diagnosed with a complex condition

might ask which physicians have the most experience treating patients with similar characteristics, and which have achieved the best outcomes. The service could analyze de-identified histories of comparable patients, relate conditions to providers, treatments, and outcomes, and return comparative information without exposing the identities or histories of other patients.



A physician considering a treatment could ask how alternative treatments have been performed for patients similar to the one being treated. AI could identify an appropriate comparison population using relevant demographic and clinical characteristics, analyze treatments and outcomes, and present the results alongside information from the individual patient's history that may affect the decision.

A researcher studying disease progression might be authorized to analyze de-identified information from patients who meet defined clinical and demographic criteria. Rather than searching individual medical records, AI could help identify the appropriate cohort and analyze conditions, observations, laboratory results, medications, treatments, and outcomes across years or decades.

A pharmaceutical company investigating a potential adverse effect might need information about patients who received a particular medication, their relevant characteristics, dosage and duration, other medications, subsequent clinical events, laboratory results, and outcomes. The analysis could identify patterns and populations warranting further investigation without disclosing patient identities.

A medical-device manufacturer evaluating an implant might analyze device type, implantation date, relevant patient characteristics, subsequent procedures, complications, revisions, and outcomes. AI could identify patterns associated with device longevity or failure while the manufacturer receives only the information or aggregated results authorized for that purpose.

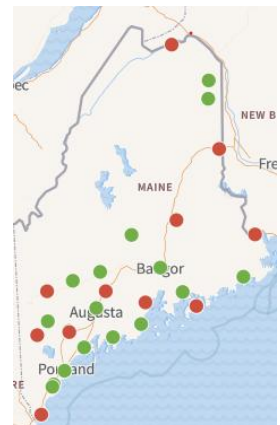
A payer evaluating a prior authorization request could use AI to assemble the relevant clinical evidence from the longitudinal history and determine whether the request satisfies established criteria, allowing appropriate requests to be approved quickly with little or no additional provider documentation. More complex requests could be evaluated using the patient's broader clinical history, previous treatments and outcomes, risk factors, the requesting provider's relevant experience, and outcomes for comparable patients. This is particularly important when a patient's history makes a routine authorization requirement inappropriate. Rather than simply reducing the cost of prior authorization, better information could reduce delays in necessary care and identify situations in which denying or delaying a test or treatment ultimately produces worse outcomes and greater expense.

The longitudinal history could also **make payer performance measurable**. Authorization requests, approvals, denials, delays, appeals, subsequent treatment, patient outcomes, and total costs could be analyzed together to determine whether payer policies actually reduce unnecessary care or merely postpone necessary care and increase later costs. Aggregated results

could identify unusually high denial rates, repeated requirements for treatments that prove ineffective, differences in approval times, and authorization practices associated with poorer outcomes. Such information could help regulators evaluate payer practices and eventually help employers and individuals consider payer performance when selecting health coverage.

A government agency investigating potential healthcare fraud could use claims, providers, patients, procedures, diagnoses, payments, clinical events, and other supporting information to identify unusual patterns that warrant investigation. AI could look for relationships and anomalies that are difficult to detect within a single payer's claims—for example, improbable procedure volumes, billing inconsistent with documented care, or patterns involving multiple providers and payers. Initial analysis could use de-identified information, with access to identified information requiring additional authorization when justified for an investigation.

Public-health planners evaluating access to healthcare could analyze the geographic distribution of patients, conditions, providers, specialties, services, and utilization to identify communities where available healthcare capacity does not match population needs. AI could identify areas with shortages of particular specialties, excessive travel for treatment, long delays in obtaining care, or growing demand associated with demographic and disease trends. The analysis could support decisions about where additional services are needed without exposing individual patient identities.



Current/Former birthing centers in Maine

These requests use the same longitudinal information resource, but they do not use the same information or expose the same level of detail. Some require identified information about a specific patient. Others require de-identified information across populations. Still others may require only an aggregate result.

AI makes these capabilities practical because the relevant information may span millions of patients, decades of history, multiple types of clinical information, and complex relationships among conditions, treatments, providers, devices, and outcomes. The architecture constrains AI to the

information authorized for the user and purpose, while AI helps determine what within that authorized information is relevant and how to analyze it.

The objective is not to provide access to more data. It is to provide the right information, for the right purpose, to the right authorized user.

Conclusion — One History, Many Uses

The value of a longitudinal patient history extends far beyond giving a physician better access to a patient's medical records. The same information, securely accessed for different authorized purposes, can improve decisions throughout healthcare.

For patients, it can mean finding providers with the most relevant experience, understanding treatment alternatives, avoiding treatments that previously failed, and receiving care based on a more complete understanding of their history. For physicians, it can mean having the relevant information available when a decision must be made rather than searching through fragmented records or relying on what the patient can remember.

Across millions of patients, the same information becomes an extraordinarily valuable source of evidence. It can help determine which treatments work best for which patients, identify differences in provider experience and outcomes, detect drug and device problems earlier, expose patterns of fraud and unnecessary care, identify communities where healthcare services do not match population needs, and accelerate research that today depends on fragmented and incomplete information.

These benefits reinforce one another. Better information can lead to better treatment decisions. Measuring the results of those decisions can identify more effective treatments and providers. Applying that knowledge to future decisions can improve outcomes while avoiding ineffective care. Earlier detection of adverse drug effects, device failures, disease trends, and gaps in healthcare access can prevent problems rather than pay for their consequences.

The economic opportunity is equally broad. Savings do not have to come from a single dramatic change to healthcare. They can accumulate through less administrative work, fewer unnecessary tests and procedures, faster prior authorization, reduced fraud, fewer avoidable complications, better

selection of treatments and providers, earlier identification of ineffective drugs and devices, and a better understanding of which healthcare expenditures improve outcomes. In a healthcare system spending trillions of dollars each year, improvements across many of these areas can have substantial cumulative value.

AI makes it practical to use this volume and complexity of longitudinal information, but AI is not the underlying asset. The asset is a trustworthy history connecting patients, conditions, providers, treatments, costs, and outcomes across organizations and over time. AI helps retrieve, relate, analyze, and explain that information within the authorization and privacy constraints established by the architecture.

The ultimate measure of this architecture is therefore not how much healthcare data it collects or how sophisticated its AI becomes. It is whether people **live longer and healthier lives, receive more effective care from better-informed providers, avoid treatments that do not work, and spend less—individually and collectively—for the healthcare outcomes they achieve.**

One history. Many uses. Better decisions. Better outcomes. Lower cost.

AI in the Healthcare Information Architecture

Transforming Data into Information, Knowledge, and Insight

AI in healthcare is often portrayed as taking patient information and producing a diagnosis or treatment recommendation. This chapter proposes a different role: using AI to help manage the growing volume and complexity of healthcare information. It examines how AI can extract clinical information from healthcare data, reconcile it into an evidence-linked Longitudinal Patient History, and analyze authorized knowledge to produce useful insight for providers, payers, patients, and other healthcare purposes—while preserving the evidence needed for human decisions.



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Introduction

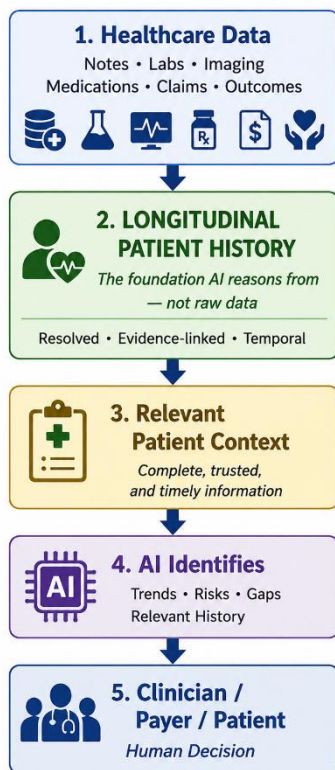
A common vision for AI in healthcare is deceptively simple: give an AI system everything known about a patient and ask it what is wrong and what treatment or medication to prescribe. That vision focuses attention on replacing or automating clinical judgment. It overlooks a more immediate problem: healthcare already generates more information than the people making healthcare decisions have the time to find, reconcile, analyze, and understand.

A national longitudinal patient history makes far more of that information available, but availability alone does not make it useful. Decades of clinical notes, diagnoses, laboratory results, imaging, medications, procedures, treatments, claims, and outcomes cannot simply be placed in front of a physician during a fifteen- or twenty-minute encounter. Nor can every payer, pharmacist, specialist, or other healthcare participant be expected to manually reconstruct the portions of that history relevant to each decision.

The role of AI proposed in the Future Healthcare Information Architecture is therefore not simply **information in, medical decision out**. AI is used throughout the architecture to help transform **Data → Information →**

Knowledge → Insight, providing the foundation for a Human Decision. It can extract clinical information from healthcare data, reconcile that information across sources and time, identify what is significant for a particular healthcare purpose, and present the resulting insight to the person responsible for making the decision.

The objective is not to ask AI to replace healthcare professionals. It is to use AI to perform information-processing work that increasingly exceeds the

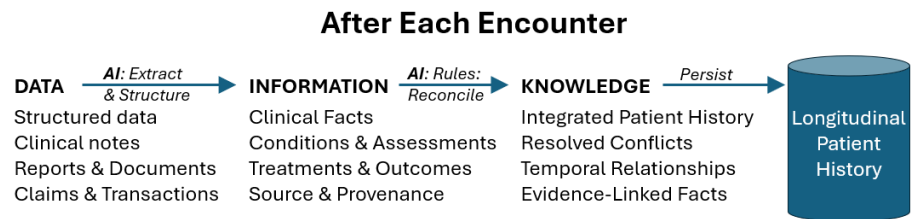


time humans can reasonably devote to it—while preserving the evidence they need to understand, verify, and ultimately decide what to do.

Building and Using Longitudinal Knowledge

From Healthcare Data to Longitudinal Knowledge

Every healthcare encounter creates new data. Some data is captured as structured data in Electronic Health Records, laboratory, imaging, pharmacy, payer, and other Healthcare Information Systems. Other information is contained in clinical notes, reports, forms, and documents written primarily for people rather than computers. The Future Healthcare Information Architecture uses both forms to continuously build and maintain the patient's Longitudinal Patient History.



The first two transformations occur after new healthcare data becomes available. Their purpose is not to make a clinical decision, but to make what happened during the encounter understandable in the context of everything previously known about the patient.

Data → Information. Structured data can often be interpreted directly, while AI can extract clinically meaningful facts from unstructured clinical notes and reports. Conditions, assessments, treatments, outcomes, and other facts are identified and structured while preserving their source and provenance. The objective is to capture what each source says without losing the evidence from which it was derived.

Information → Knowledge. New information is then integrated with information accumulated across previous encounters and healthcare organizations. Repeated information can be consolidated, conflicting information identified and reconciled, and relationships across time established. The result is an evidence-linked understanding of the patient's history that is persisted in the Longitudinal Patient History while retaining the underlying information needed to verify how that knowledge was derived.

These transformations require AI and deterministic processing to perform specific tasks consistently according to defined rules. We discuss how those capabilities are developed, constrained, and validated later.

From Longitudinal Knowledge to Insight

The Longitudinal Patient History may eventually contain decades of knowledge assembled from encounters across many healthcare organizations. Making all that knowledge available does not mean presenting it for every purpose. A physician preparing to see a patient, a payer reviewing a prior authorization, and an analyst evaluating treatment effectiveness require different portions of the same longitudinal history.



A library of standardized retrieval requests can provide detailed or summarized views of the Longitudinal Patient History for different purposes. Authentication identifies who is making the request, while privileges determine what that individual or organization is permitted to retrieve. The result is a purpose-specific set of longitudinal knowledge rather than the patient's entire history.

Knowledge → Insight. AI analyzes the retrieved patient knowledge, together with relevant external knowledge when required, to determine what is significant for the particular purpose. For a provider encounter, this might include important findings from the patient's history, trends and changes over time, risks or gaps requiring attention, and possible treatment options informed by clinical guidelines or evidence. The objective is not simply to summarize what was retrieved, but to identify what deserves the provider's attention and preserve the evidence supporting those observations.

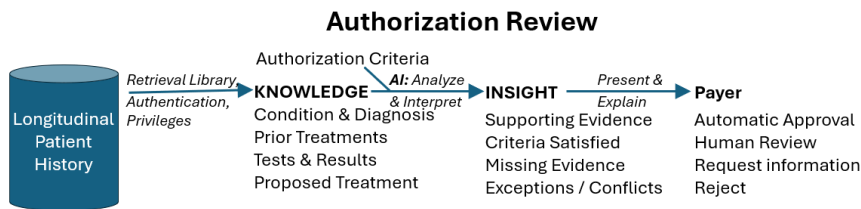
Insight → Human Decision. The resulting insights are presented to the provider before or during the encounter. The provider can examine the supporting evidence, consider information obtained directly from the patient, discuss alternatives with the patient, and decide what action is

appropriate. AI reduces the information-processing burden; the healthcare professional remains responsible for the clinical decision.

One History, Multiple Uses

The same Longitudinal Patient History can support many different participants and purposes without creating a separate patient history for each application. Providers, payers, patients, researchers, public health organizations, and government agencies may draw upon the same underlying longitudinal knowledge. Still, they do not need the same information or use it in the same way.

The retrieval library provides purpose-specific views of the history, while authentication and privileges control what each requester may access. AI can then analyze the retrieved patient knowledge for the particular task, combining it when necessary with external knowledge, rules, or criteria appropriate to that use. The resulting insight is therefore specific to both the patient and the purpose for which the knowledge was requested.



Prior authorization provides one example. A payer could retrieve the portions of the patient's history relevant to a requested treatment and compare that evidence with its authorization criteria. Rather than requiring provider and payer personnel to repeatedly assemble and review supporting documentation, AI could identify which requirements are supported by the longitudinal history, where the supporting evidence resides, and what information may still be missing. The payer's authorization criteria remain separate from the patient's history, while the Longitudinal Patient History provides the evidence needed to apply them.

Examples of Retrieval Library Requests

Content Retrieval Services can support many healthcare questions using a library of standardized retrieval requests. Each request follows the same

basic process: retrieve the longitudinal knowledge relevant to a particular purpose, analyze it to produce insight, and return only the information the requester is authorized to receive. The examples below illustrate potential requests, the knowledge that might be retrieved, how AI could analyze it, and whether the resulting insight requires an identified patient, de-identified patient histories, or only summary information.

Examples Of How Longitudinal Knowledge Can Be Used

Purpose / Question	Knowledge Retrieved	AI Analysis	Insight / Privacy
What treatments have already been tried?	Condition, treatments, dates, response, side effects, outcomes	Relate treatments and outcomes across encounters; identify unsuccessful or discontinued treatments	Patient identified: treatment history relevant to the current condition
How have test results changed over time?	Test type, results, reference ranges, dates, related conditions and treatments	Normalize comparable results; identify trends, significant changes and anomalies	Patient identified: trends and changes with supporting results
Which treatment works best for patients like this?	Condition, patient characteristics, treatments, outcomes, complications	Find comparable patients; compare outcomes among treatment alternatives	Patient de-identified: treatment effectiveness summarized for comparable patients
Which provider has the most experience—and best outcomes?	Conditions treated, procedures/treatments, patient risk factors, outcomes, provider	Risk-adjust provider experience and outcomes for comparable patients	Patients de-identified; providers identified: provider-level comparative results

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Examples Of How Longitudinal Knowledge Can Be Used

Purpose / Question	Knowledge Retrieved	AI Analysis	Insight / Privacy
Should this test be approved?	Condition, symptoms, prior tests, treatments, outcomes, proposed test	Compare patient evidence with payer authorization criteria	Patient identified: criteria satisfied, supporting evidence and missing requirements
Which payer policies cause harmful delays?	Authorization requests, criteria, approvals/denials, treatment timing, subsequent outcomes	Compare delays and outcomes associated with authorization policies	Patients de-identified: summary results by policy/payer
Is this drug causing unexpected problems?	Medication, dosage, duration, patient characteristics, conditions, adverse outcomes	Identify outcome patterns associated with the drug and potentially affected populations	Patients de-identified: aggregate adverse-outcome patterns
Where are providers needed?	Patient demographics/geography, conditions, referrals, provider specialty, travel/access patterns	Identify geographic demand, shortages and access gaps	Summary data: geographic need without patient identification

How AI Is Used in the Architecture

The preceding examples illustrate that AI is not a single step between healthcare data and a decision. It performs different information-processing tasks at different points in the architecture. After an encounter, AI helps extract and structure clinical information from newly received data and reconcile that information with the patient's existing longitudinal history. When that history is later used, AI analyzes purpose-specific knowledge to identify the findings, relationships, trends, risks, or options that deserve attention. These three uses of AI—**Extract & Structure, Reconcile & Integrate, and Analyze & Interpret**—require different instructions, rules, knowledge, and methods for determining whether the results are reliable.

Extract & Structure: Data → Information

Healthcare data arrives in many forms. Some information is already structured—laboratory results, prescriptions, procedure codes, claims, vital signs, and other discrete fields can often be mapped directly into the longitudinal data model. Much of the clinical meaning, however, remains embedded in physician notes, imaging reports, discharge summaries, referral documents, and other narrative text. AI, particularly Large Language Models (LLMs), can help transform this unstructured data into structured clinical information.

Patient reports right shoulder pain for six weeks. A rotator cuff injury 10 years ago was treated with physical therapy. Examination shows limited range of motion and pain with abduction. Possible rotator cuff tendinopathy; a tear cannot be excluded. Acetaminophen has provided limited relief. Physical therapy is recommended. MRI will be considered if symptoms persist

Legend
Condition/Finding
Past Treatment
Future Treatment

The task is more difficult than identifying medical terms in text. A clinical note may mention a condition because the physician diagnosed it, the patient reported having it years ago, a family member had it, it is being considered as a possible diagnosis, or the physician specifically ruled it out. Similarly, a treatment may have been performed, recommended, considered, discontinued, or reported by the patient as having occurred elsewhere. Extracting a term without understanding its context could turn what the note actually says into incorrect patient information.

The AI must therefore be taught what information to extract and how to classify what it finds. For each relevant statement, it may need to identify the clinical concept, its status, when it occurred, how it relates to the current encounter, and whether it represents a physician assessment, patient statement, test result, treatment, outcome, or other type of evidence. The output should conform to the defined longitudinal data model rather than allowing the AI to invent its own representation.

Just as important, the transformation must preserve provenance. An extracted fact should remain linked to the note, report, or structured record from which it was derived. The objective is not merely to create more structured data; it is to create clinical information whose meaning and source can later be evaluated, reconciled, and, when necessary, verified against the original evidence. When an extraction is found to be incorrect, that correction can also serve as feedback to improve the extraction process and reduce similar errors in the future.

A related use of AI is already being applied in public health. CDC's One CDC Data Platform¹¹ provides shared capabilities for data ingestion, preparation and cleaning, analytics, and dissemination. One application uses AI to extract, standardize, categorize, and analyze information for foodborne-outbreak investigations—work CDC reports can reduce from hours to less than ten minutes. **Most information in the platform is de-identified, and its public-health purpose is substantially different from maintaining an individual longitudinal patient history.** But the example demonstrates how AI can help transform incoming data into structured, usable information before analysis and decision-making.

Reconcile & Integrate: Information → Knowledge

Extracting information from each healthcare encounter does not by itself create a longitudinal understanding of the patient. The same condition may be described differently by multiple providers, appear repeatedly over many

¹¹ CDC, “One CDC Data Platform.” Describes 1CDP's shared capabilities for data ingestion, preparation and cleaning, analytics and visualization, and dissemination and reporting; also describes AI-assisted extraction, standardization, categorization, and analysis for foodborne outbreak investigations.

<https://www.cdc.gov/data-modernization/php/one-cdc-data-platform/index.html>

years, change in severity or status, or be contradicted by later information. Treatments may occur across different organizations, and their outcomes may not be documented until months or years later. The architecture must relate each new piece of information to what is already known about the patient.

AI can help determine whether newly received information confirms, updates, supplements, or conflicts with existing information. A new diagnosis may establish a previously unknown condition, confirm an existing condition, refine its description, indicate that it has resolved, or contradict an earlier assessment. Similarly, a treatment must be related to the condition it was intended to address, and

subsequent results, complications, or side effects must be associated with that treatment when the available evidence supports the relationship.

Reconciliation cannot simply assume that the newest information is correct. A patient's answer on a current intake form may conflict with years of clinical evidence, while a recent diagnostic test may legitimately overturn an earlier diagnosis. AI must therefore apply defined rules for considering the source, type, timing, and strength of the available evidence. Where the evidence does not support a definitive conclusion, the uncertainty or conflict should remain visible rather than being resolved arbitrarily.

The result is knowledge that represents what the accumulated evidence establishes about the patient across organizations and time. The original information and its provenance remain available, allowing a derived conclusion to be traced back to its supporting or conflicting evidence. In this way, the Longitudinal Patient History becomes more than a collection of records: it maintains an evidence-linked understanding of conditions, treatments, outcomes, and their relationships as that understanding changes over time.

INFORMATION



2016 — Patient History
Rotator cuff tear



2026 — PCP Assessment
Possible rotator cuff tendinopathy



2026 — MRI
No tear; tendinopathy



AI + Rules: Reconcile & Integrate



KNOWLEDGE



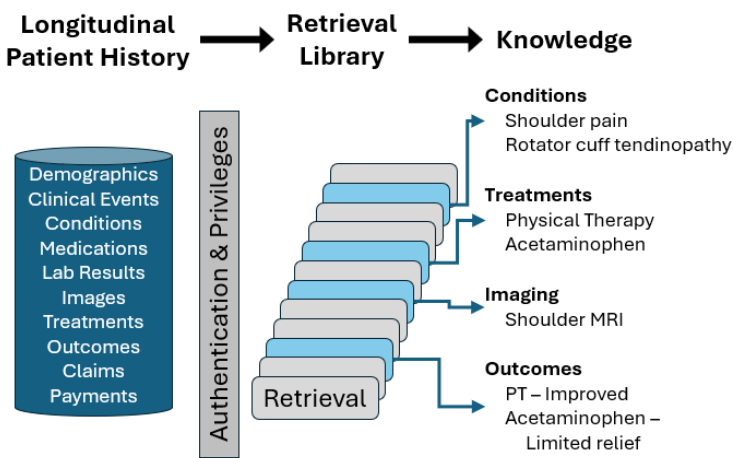
Rotator Cuff Tendinopathy — Current



Prior tear — Patient reported

Retrieving Knowledge for a Purpose

Before AI can analyze knowledge from the Longitudinal Patient History, it must retrieve the knowledge appropriate to the purpose. Rather than giving an AI system unrestricted access to everything known about a patient, the architecture can provide a library of standardized retrieval services designed for different healthcare uses. A request might retrieve conditions and their history, treatments and outcomes, medications, laboratory trends, prior procedures, or other defined combinations of longitudinal knowledge. Depending on the purpose, the service could return detailed records, summarized knowledge, or knowledge covering a specified time period.

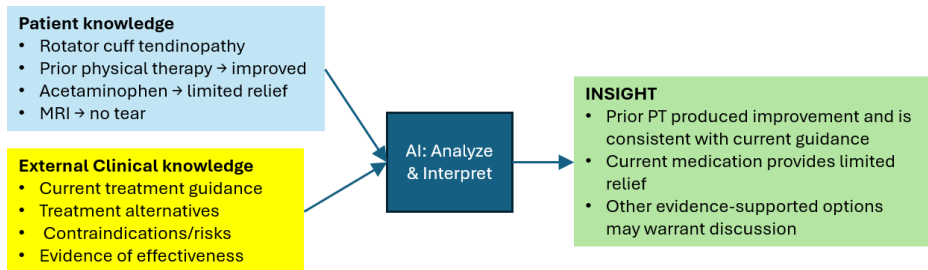


Authentication identifies the individual or system making the request, while privileges determine which knowledge the requester is authorized to receive. The same retrieval service could therefore produce different results depending on the requester and purpose, including identified patient information, de-identified information, or aggregate results. These controls separate access to longitudinal knowledge from the AI that subsequently analyzes it.

Standardized retrieval services also provide AI with more predictable inputs. Instead of requiring each AI application to locate, interpret, and filter an unrestricted collection of records, the retrieval service supplies defined tables of relevant longitudinal knowledge. AI can then concentrate on the next task: analyzing that knowledge to determine what is significant for the purpose for which it was requested.

Analyze & Interpret: Knowledge → Insight

A Longitudinal Patient History may contain decades of knowledge, but its value depends on determining what matters for a particular purpose. Standardized retrieval services provide the authorized patient knowledge appropriate to the task. Analysis may also combine that patient-specific knowledge with clinical guidelines, treatment alternatives, evidence of effectiveness, or other knowledge relevant to the purpose.



AI can then analyze the retrieved knowledge to identify relationships, trends, changes, risks, gaps, or other findings that might otherwise require substantial human effort to discover. For a provider encounter, this could include recognizing that a condition is worsening, identifying a previously ineffective treatment, detecting a potential conflict with another condition or medication, or highlighting information that may warrant further investigation. The objective is not to repeat or summarize everything retrieved, but to identify what is significant enough to deserve attention.

The analysis may also require knowledge beyond the individual patient's history. Evaluating treatment options could combine the patient's conditions and prior treatments with outcomes and costs observed among comparable patients. Prior authorization could compare patient-specific evidence with payer criteria. Other uses may incorporate clinical guidelines, policies, population experience, or other information appropriate to the question being addressed.

What constitutes useful insight therefore depends on the purpose. AI must be taught what to look for, what additional knowledge or rules to apply, and how to distinguish significant findings from information that is merely present. The resulting insight should identify the evidence supporting it and communicate uncertainty where appropriate, allowing the healthcare

professional or other authorized user to understand why something was highlighted and determine what action, if any, should be taken.

Making AI Reliable for Healthcare Use

The AI functions described in this architecture would require substantial development and validation before they could be relied upon for healthcare use. A capable Large Language Model is only a starting point. Each function must precisely define what the AI is expected to do, establish how healthcare information and conflicting evidence should be interpreted, provide the knowledge and rules required for the task, and demonstrate through extensive testing that the results are sufficiently accurate and reliable for their intended use. This would require continuing collaboration among clinicians, healthcare organizations, payers, information architects, AI specialists, and other domain experts.

Instructions and definitions can establish the meaning of the information being processed and the expected output. For clinical information extraction, these instructions might define how to distinguish a confirmed diagnosis from a suspected condition, patient-reported history, family history, or a ruled-out condition. They can also specify how extracted information maps to the longitudinal data model and what provenance to retain.

Representative Cases can teach the AI how to apply those definitions to realistic healthcare information. Representative clinical notes can be paired with the facts and relationships that should be extracted. More difficult examples can demonstrate how to handle ambiguity, conflicting statements, incomplete information, or terminology that varies among providers. Similar examples can be developed for reconciliation, treatment analysis, prior authorization, and other defined functions.

Rules can constrain decisions that should not be left solely to an AI model's judgment. Reconciliation rules might specify how to consider the source, timing, and strength of evidence when information conflicts. Prior-authorization rules can define the criteria that must be satisfied. Other functions may combine deterministic processing with AI interpretation so that established healthcare or business rules are applied consistently.

External knowledge may also be required. The Longitudinal Patient History describes what is known about an individual patient, but some questions require information beyond that history. Treatment analysis might use clinical guidelines and outcomes from comparable patients. Prior authorization might require the payer's criteria for pre-approval of the proposed treatment or procedure. Other applications may require policies, reference information, cost data, or population-level evidence appropriate to the task.

Validation must establish that each AI function performs reliably across different conditions, providers, terminology, documentation styles, patient populations, and unusual or conflicting situations. Testing must include not only common cases, but cases specifically designed to expose incorrect inference, missing information, unsupported conclusions, and other potentially consequential errors.

The objective is therefore not to train a single AI system to "understand healthcare." Instead, it defines specific AI functions within the architecture and provides each with the instructions, examples, rules, knowledge, and validation appropriate to its purpose. As AI models improve, you can replace or upgrade individual models without changing the fundamental information architecture or the responsibilities assigned to each function.

Conclusion

The opportunity for AI in healthcare is broader than giving an AI system a patient's records and asking it to recommend a diagnosis or treatment. Healthcare already generates more information than participants can reasonably find, reconcile, analyze, and understand. A national Longitudinal Patient History makes more information available; AI can help make it usable.

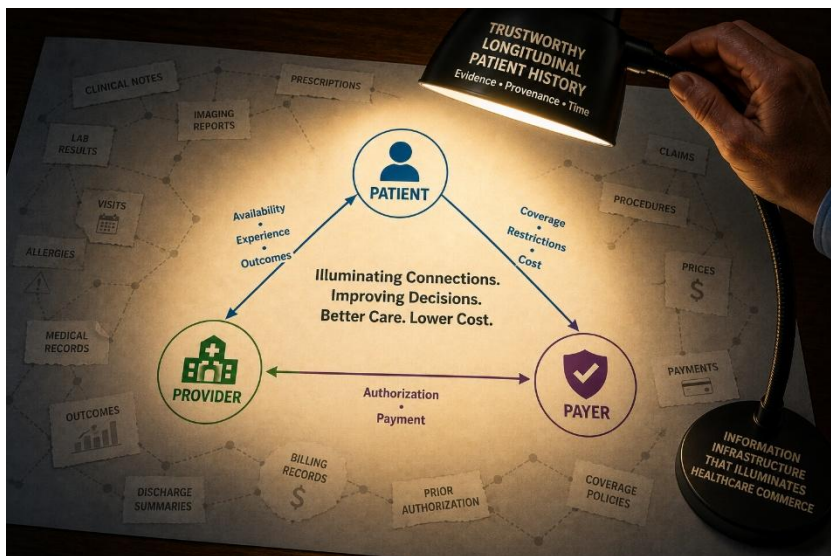
Within the Future Healthcare Information Architecture, AI performs specific functions: extracting clinical information from data, reconciling it into longitudinal knowledge, and analyzing authorized knowledge to produce insight for a particular purpose. These functions can be governed by defined instructions, rules, external knowledge, and validation, while preserving the evidence supporting their results.

The objective is not to replace human judgment with an automated answer. It is to use AI to transform **Data → Information → Knowledge → Insight**, giving healthcare professionals and other authorized participants better information for **human decision-making**.

Bringing Healthcare Commerce into the Light

Transforming Data into Information, Knowledge, and Insight

Healthcare decisions are often made with remarkably limited information about alternatives, costs, experience, and outcomes. This chapter examines how a national longitudinal patient history could bring healthcare commerce into the light—giving patients, providers, payers, and government better information while enabling informed choice and competition to improve access, effectiveness, and value.



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Introduction

Recently, I developed severe neck pain that made it difficult to look left, right, or up. I called my doctor's office and was told they could get me an appointment with a spine specialist—in about a month. That is a long time to endure pain and limit normal activities. Were there other specialists who could see me sooner? How would I find them? And how would I know whether they were good choices?

A friend needed to reschedule a video appointment with a nurse practitioner because of another important medical appointment. The next available video appointment was almost two months later. Was another provider available sooner? There was no easy way to know.

I read an account on Facebook from a woman with a lump on her chest that was growing by the hour and becoming hot. One emergency room prescribed a low-dose antibiotic, heat, and a mammogram. The next day, she continued to worsen; she went to a different ER that performed blood tests and an ultrasound and gave high-dose antibiotics intravenously. She was told she had severe cellulitis that could have progressed to deadly sepsis.

These are anecdotes and cannot establish whether the care provided was clinically appropriate. But they illustrate a larger problem: **healthcare consumers often make important decisions with remarkably little information about their alternatives.** Who is available? What will it cost? What experience does the provider have? What results have other patients experienced?

Even when a physician recommends a test or treatment, a payer may require prior authorization, additional evidence, or that a less expensive treatment be tried first.

Imagine buying a car this way. You may not know its ultimate cost before committing to buy it, have little information about its reliability, and have someone else paying most of the bill determine which options you can purchase. We would hardly expect such a marketplace to produce vigorous competition based on price and performance.



Healthcare isn't car buying. Medical decisions require specialized knowledge, emergencies can eliminate choice, and outcomes vary among patients. But those differences don't explain why information about healthcare choices should remain so difficult to obtain.

Bringing healthcare commerce into the light means giving consumers better information about availability, cost, experience, and observed outcomes. Freedom of choice becomes far more powerful when people have the information necessary to exercise it.

Healthcare Is a Market Without Normal Market Information

In most markets, consumers expect information before making a purchase. They can compare products, prices, availability, features, warranties, and other buyers' experiences. Competition gives suppliers an incentive to offer better products, better service, or lower prices.

Healthcare operates differently. Patients may be able to choose among physicians, hospitals, imaging centers, laboratories, insurance plans, and other healthcare organizations. Still, much of the information needed to make those choices is difficult or impossible to obtain.

Consumers have attempted to fill the information vacuum themselves. They ask friends and neighbors, post questions to community Facebook groups, and consult online provider ratings. These sources can help you

learn whether a physician communicates well, the staff is helpful, or appointments run on time. But a five-star review tells us remarkably little about what may matter most: How much experience does this provider have treating patients like me? What treatment was provided? What did it cost? And what happened to those patients afterward?



Even a review saying, "Dr. Jones fixed my shoulder," provides limited evidence. Was that patient's shoulder problem the same as mine? Were we similar in age and health? Had we tried the same previous treatments? Would the patient have improved regardless? Patient reviews can describe an experience, but they rarely provide enough information to compare healthcare effectiveness.

Price can be equally difficult to understand. The amount billed may bear little relationship to the amount accepted from a payer, and what the patient ultimately pays depends on negotiated rates, deductibles, copayments, coinsurance, network status, and the services that follow. The least expensive initial treatment may not be the least expensive course of care if it fails and additional treatment becomes necessary.

The opacity extends beyond providers. Consumers also choose payers, either directly or from alternatives offered by an employer or government program. They can compare premiums, deductibles, copayments, and provider networks, but it can be much harder to understand how the payer may constrain future healthcare choices.

What diagnostic procedures require prior authorization? What treatments must be tried before another will be approved? How often are authorization requests denied? How long does approval typically take? How restrictive is the



provider network when specialized care is needed? These differences may be relatively unimportant when selecting an insurance plan but become extremely important after someone becomes ill.

Healthcare transparency therefore needs to mean more than publishing prices. A transparent healthcare marketplace would make the information that materially affects a healthcare decision visible **before the decision must be made.**

Today, without that transparency, patients may choose providers based on referrals, recommendations from friends and neighbors, online reviews, reputation, location, insurance acceptance, or simply who can give them an appointment. Those sources can be useful, but they provide little objective information about the provider's experience, cost, or results with patients like them.

What patients need when choosing a provider is better information about:

- **Availability:** Who can provide the care, where, and how soon?
- **Experience:** How much experience does the provider have treating patients with similar conditions and circumstances?
- **Cost:** What is the likely cost to the patient and payer—not just for the initial service, but for the course of care?
- **Observed outcomes:** What happened afterward to comparable patients receiving similar care?

The same problem exists when choosing a payer. Today, consumers commonly compare premiums, deductibles, copayments, provider networks, and covered benefits. What may be much less apparent is how the payer will affect their choices after they become ill and need care.

What consumers need when choosing a payer is better information about:

- **Provider access:** How broad is the network, particularly for specialists, and how readily can members obtain needed care?
- **Prior authorization:** Which tests, treatments, and medications require approval before they can be provided?
- **Restrictions:** What treatments must be tried first, and what other requirements can limit the care recommended by a physician?
- **Approval and denial experience:** How often are requested services approved or denied, and how long do those decisions typically take?
- **Appeals:** How frequently are denials reversed, and how long does the appeal process take?

None of this information can guarantee a successful outcome or identify a single "best" provider or payer. Healthcare is too complex, and patients and their circumstances are too different, for such certainty. But uncertainty does not make information worthless.

A functioning market does not require perfect information. It requires enough information for buyers to make meaningful choices and for suppliers to compete for those choices. Healthcare frequently provides neither.

What Happens When Healthcare Has to Compete?

We do not have to rely entirely on theory to consider what greater consumer choice might do to healthcare. Some healthcare services operate much more like conventional markets because consumers pay directly and can choose among competing providers.

LASIK eye surgery provides an interesting example. Because LASIK is generally elective, non-urgent, and paid directly by the consumer rather than a third-party payer, prospective patients have both the opportunity and financial incentive to compare providers. They can ask what the procedure will cost, investigate alternatives, and decide whether and where to have it performed.

Research presented to Congress found that competition helped keep LASIK prices down. The inflation-adjusted price of conventional LASIK declined nearly 30 percent over roughly a decade. Researchers attributed the competitive environment partly to consumers' ability to obtain price information relatively easily and to ophthalmologists' ability to enter the market and compete for patients.¹²

¹² **U.S. House Committee on Ways and Means, Subcommittee on Health, *Price Transparency*, July 18, 2006.** Testimony of Ha T. Tu, Center for Studying Health System Change. Research on the LASIK market found that the inflation-adjusted average price of conventional LASIK declined nearly 30 percent over the preceding decade. The testimony attributed the market's price competitiveness largely to relatively easy entry by ophthalmologists and consumers' ability to obtain price quotes at little cost or inconvenience.
<https://www.govinfo.gov/content/pkg/CHRG-109hrg30452/pdf/CHRG-109hrg30452.pdf>

LASIK also demonstrates that competition alone does not solve the information problem. Researchers found that providers did not always package services consistently, making direct price comparisons difficult. Quality varied among providers, and consumers continued to rely heavily on word-of-mouth recommendations. Some advertised prices could also be misleading.

That distinction is important. The lesson from LASIK is not that every healthcare service should operate like an elective self-pay procedure. Emergencies cannot be comparison-shopped, many medical decisions require substantial clinical expertise, and insurance protects patients from costs they could not reasonably bear themselves.

The more useful lesson is that **when consumers have meaningful choices, useful information, and an economic reason to consider the alternatives, providers have an incentive to compete for their business.**

That competition does not have to be limited to price. Providers can compete on availability, convenience, experience, service, technology, and observed outcomes. A provider who can deliver comparable care sooner, at lower cost, or with better observed results has something meaningful to offer prospective patients—if patients can see the difference.

Much healthcare suppresses those signals. The patient may not know the price, the payer may determine which providers or treatments are available, and information about provider experience and outcomes may be unavailable. When

consumers cannot distinguish among alternatives, providers have less reason to compete on differences consumers cannot see.

Bringing healthcare commerce into the light would give consumers the information and freedom of choice needed to make competition meaningful. Providers who deliver better outcomes, greater access, or better value would have an opportunity to attract more patients. Those who do not would have



an economic incentive to improve. That is how informed choice can turn transparency into a force for improving healthcare.

Payers Are Making Decisions in the Dark Too

During a business discovery discussion years ago, an actuary made a comment that stayed with me: **“There are no bad policies, only bad rates.”** His point was that insurance ultimately depends on collecting enough money across a population to pay the claims that population will generate.

Insurance companies are not charities. They pool risk so that an individual facing a major illness does not have to bear the entire cost alone. The money to pay healthcare claims must ultimately come from premiums, employers, government programs, or patients themselves. Insurers must collect enough to pay those claims and their operating expenses and, for investor-owned companies, earn a return sufficient to attract capital.

That gives payers a legitimate reason to question whether healthcare expenditures are necessary and appropriate. Paying every claim and approving every requested treatment without scrutiny would eventually increase insurance costs for everyone.

The problem is that payers may be making those decisions with limited information.

Consider a request for an expensive MRI. Ideally, the decision would consider the patient's symptoms, clinical history, previous injuries and illnesses, prior imaging, treatments already attempted, changes in the condition, and the physician's reason for ordering the test. Without ready access to that broader clinical context, a payer may have little choice but to rely on the documentation submitted with the request and rules developed for patient populations.

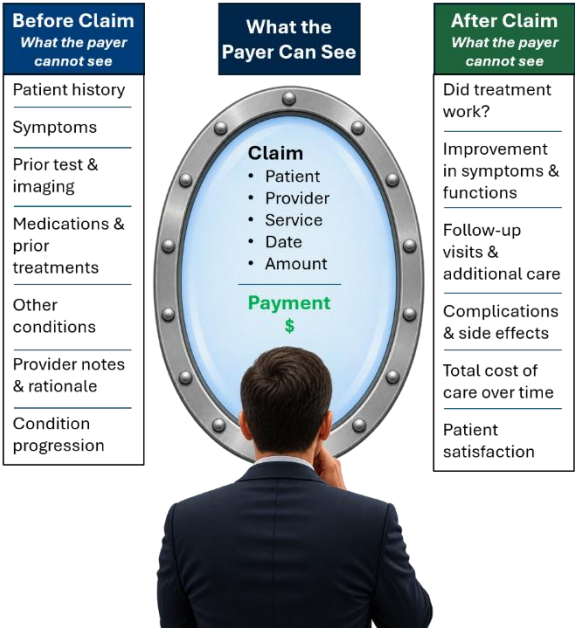
Those rules can become blunt instruments:

- Try physical therapy first.
- Obtain an X-ray before an MRI.
- Try the less expensive medication first.
- Provide additional documentation before approval.

Such requirements may be reasonable for many patients. But the treatment that usually makes sense for a population may not make sense for a particular individual. A requirement intended to avoid an unnecessary expense can delay diagnosis or treatment—and in some cases ultimately increase the total cost of care.

The same longitudinal information could strengthen fraud and improper-payment detection by allowing claims to be evaluated against a broader record of clinical events, diagnoses, procedures, and treatments rather than primarily against other claims. A suspicious claim could be examined in the context of what actually happened to the patient, making it easier to distinguish legitimate care from billing that warrants further investigation.

But the value of longitudinal information should not end when the claim is paid. It can also help answer the question that claims data alone rarely answer: **What happened to the patient afterward?** Did the patient improve? Was additional treatment required? Were there complications? And what was the total cost of care over time?



Those answers could provide evidence for improving future authorization policies. If an expensive treatment consistently produces good outcomes for patients with particular characteristics, that history should matter when a similar patient needs the same treatment. Conversely, if a less expensive treatment that is routinely required first frequently fails for certain patients and merely delays the treatment they ultimately receive, that should matter too.

Better information does not eliminate the payer's responsibility to control costs. It provides better tools to do so. The objective should not be to

approve more care or deny more care. It should be to approve appropriate care faster, detect inappropriate expenditures more accurately, learn from previous costs and outcomes, and make future decisions based on the likely total course of care rather than simply the cost of the next claim.

What Should the Federal Government Do?

The federal government faces many of the same information problems as patients, providers, and payers—but on a much larger scale. **In 2026 alone, the federal government is expected to spend roughly \$1.9 trillion on Medicare, Medicaid, and other major healthcare programs.** It also influences the supply of healthcare professionals and sets rules that affect healthcare organizations nationwide.

With that scale comes the responsibility to control costs, protect public funds from fraud, and ensure federal programs work as intended. **Without sufficiently detailed and timely information, however, government actions can become blunt instruments.**

Targeting Fraud

Recent hospice fraud investigations in California illustrate both the scale of the problem and the difficulty of responding once fraud becomes widespread. Federal authorities suspended Medicare payments to **447 hospice providers and 23 home health agencies in the Los Angeles area**, involving an estimated **\$600 million in suspected fraud**. Investigations have found schemes involving patients who were not terminally ill, people enrolled in hospice without their knowledge, services that were never provided, and the misuse of identities. The concentration itself was



447

Hospice providers suspended

\$600 MILLION

Suspected fraud

extraordinary: state records showed more than half of California's hospice providers located in Los Angeles County, and investigators found dozens of hospice businesses associated with the same office location.

This is exactly the kind of problem that better national information should expose much earlier. Instead of discovering hundreds of questionable providers after enormous amounts have been billed, analytics combining enrollment, claims, patient histories, provider locations, and clinical evidence could identify implausible patterns as they develop and focus investigation on the providers creating them.

Concern about fraud has also led the federal government to freeze funding while requiring additional documentation and review. In January 2026, it froze access to certain child-care and family-assistance funding for five states. The states sued, and a federal court blocked the freeze while litigation proceeded. **Broad actions intended to stop fraud can entangle legitimate services and funding while adding litigation costs.**

Responding to Workforce Needs

Healthcare workforce policy provides another example of the long-term consequences of broad intervention. In the 1990s, policymakers were being warned of a coming physician surplus. The Balanced Budget Act of 1997 generally limited the number of residency positions for which individual teaching hospitals could receive Medicare graduate medical education funding, using fiscal year 1996 training levels as the baseline. Nearly three decades later, an aging population is consuming more healthcare, and many communities report difficulty obtaining timely access to physicians and specialists.

Wondering why you may need to wait several months to see a specialist?

The lesson is not that government should withdraw from healthcare. Government has responsibilities that individual patients, providers, and payers cannot perform. The question is **where federal involvement provides the greatest value and where decisions are better left to patients, healthcare professionals, payers, markets, and states.**

Facilitating Interstate Commerce

The Constitution provides a useful starting point. Article I gives Congress the power **“To regulate Commerce with foreign Nations, and among the several States.”** In *Gibbons v. Ogden* in 1824, Chief Justice John Marshall described commerce among the states as commerce **“which concerns more than one state.”**

Modern healthcare clearly crosses state boundaries. A patient may live in one state, receive treatment in another, and use insurers, pharmacies, laboratories, and other organizations operating nationally. During a lifetime, that patient may move repeatedly while decades of healthcare information remain scattered among organizations in multiple states.

A national longitudinal patient history does not require the federal government to decide which physician a patient should see, what a physician should prescribe, which treatment a payer should approve, or how much every provider should be paid. Those decisions can remain with the participants responsible for making them.

The federal government can establish the infrastructure and standards that allow trustworthy healthcare information to follow patients across state and organizational boundaries, with safeguards for privacy, security, auditability, and appropriate use.

Better national information could also make government oversight more precise. Claims combined with relevant clinical information could identify implausible billing and enable earlier investigations. Information about provider supply and demand could similarly reveal developing shortages before broad corrective policies become necessary.

The objective is not to replace state authority, private healthcare organizations, insurance companies, or individual choice with a national

The Interstate Highway System Facilitates Interstate Commerce



**The government built the infrastructure.
It doesn't drive the trucks or cars.**

healthcare system. It is to provide infrastructure for a healthcare marketplace that already operates across state boundaries.

The federal role should be to bring healthcare commerce into the light—not to make every healthcare decision. National infrastructure can make information visible and trustworthy. Patients, providers, payers, states, and markets can then use that information to make better decisions.

Conclusion

Healthcare does not lack information. It cannot reliably bring together the right information about a patient, resolve inconsistencies, protect it, and make it available when and where it can improve a healthcare decision.

A national longitudinal patient history could assemble information now scattered across healthcare organizations. Common rules could determine who may see what information and for what purpose, while national access monitoring could record who accessed it, when, and why—protections difficult to achieve consistently across thousands of independent systems.

The value, however, comes from what we can learn and do with that information. **Patients** could make better-informed choices based on availability, experience, cost, and observed outcomes. **Payers** could evaluate requests with greater clinical context, detect inappropriate expenditures, and learn whether expensive treatments ultimately produced better outcomes and lower total costs. **Providers** could spend less time reconstructing patient histories, have relevant information available before an encounter, and draw on nationwide evidence about treatments and outcomes for similar patients.

None of this requires replacing today's healthcare systems or transferring healthcare decisions to the federal government. It requires an information infrastructure that connects what those systems already know, protects how that knowledge is used, and makes the results visible to those responsible for making decisions.

Bringing healthcare commerce into the light means giving patients, providers, and payers better information and greater opportunity to learn from what happened before. Better information enables better choices. Better choices create pressure for better care, better outcomes, and better value.

From Vision to Action: The Proposal

A 60-Day Feasibility Assessment

The preceding chapters describe an ambitious proposal: a national healthcare information architecture that could reduce costs, improve decisions, and make healthcare information more useful. The scale of the opportunity is substantial, but so are the technical, economic, legal, and political questions. The appropriate next step is not implementation. It is a focused, 60-day assessment to determine whether the potential benefits justify further action.



FEASIBILITY ASSESSMENT

DURATION: 60 DAYS

TEAM: 6 EXPERTS

Prescription:



DETERMINE:

1 Is it technically feasible?

2 Is the potential value significant?

3 What are the major barriers?



Prescribed by:

The Future of Better Healthcare

Date:

___/___/___



NEXT STEP: EVIDENCE BEFORE IMPLEMENTATION

From Vision to Action: The Proposal

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Purpose of the Assessment

The proposal described in this book is deliberately ambitious. It suggests that a national longitudinal healthcare information architecture could make existing healthcare information more useful—reducing administrative effort, improving decisions, and creating new opportunities to understand the cost and effectiveness of healthcare.

The next step should not be implementation. Nor should it begin with a large committee, procurement effort, or technology demonstration.

It should begin by answering three questions:

1. **Could better healthcare information create substantial measurable value?**
2. **Is an architecture capable of delivering that information technically feasible?**
3. **What are the most significant barriers to making it work?**

A focused 60-day assessment by six people with complementary expertise could provide enough evidence to determine whether the concept warrants further investigation.

The six-person team should combine expertise in healthcare delivery and clinical workflow; payer operations, claims, and payment; healthcare information systems and interoperability; data architecture and AI; privacy, security, and healthcare law; and healthcare economics and public policy. The team need not represent every healthcare constituency. Its role is to conduct the assessment, gather evidence from a much broader range of participants, and determine whether the opportunities identified are valuable, technically feasible, and reasonably addressable. Where specialized knowledge is required, the team should obtain it through the interviews rather than expanding the core assessment team.

Start With the Opportunity, Not the Architecture

The preceding chapters present one possible architecture. The feasibility assessment should not begin by asking healthcare experts whether they agree with it.

Doing so would constrain the discussion to the proposed solution.

Instead, the assessment should first discover what knowledgeable people throughout healthcare would do **if information limitations that constrain them today could be overcome.**

Healthcare professionals routinely work around fragmented, incomplete, delayed, difficult-to-interpret, or unavailable information. Some of those limitations create visible problems. Others consume substantial time and effort and have become accepted as part of delivering or paying for healthcare. Still others prevent potentially valuable things from being done at all.

The assessment should identify those opportunities before determining whether the proposed architecture—or some variation of it—could address them.

Prepare Interviewees to Think Beyond Today's Constraints

Interviewees should receive a short description of the assessment several days before the discussion. It should explain the objective without describing the proposed architecture in enough detail to influence their answers.

They might be told:

We are exploring whether better use of healthcare information could materially improve healthcare decisions, operations, costs, and outcomes.

Before the discussion, think about your own area of healthcare—not about a particular technology or proposed solution. Consider the important problems you encounter, opportunities that cannot be pursued today, and activities that consume substantial time or effort because the information needed for a decision is difficult to obtain, interpret, or trust.

Then imagine those information limitations did not exist. If the information or knowledge you wished for were available, trustworthy, and timely, what could you do differently?

Don't limit your thinking to what today's systems can provide.

The objective is to get interviewees to step outside current processes and systems long enough to consider what might be possible.

Structured Discovery Interviews

The assessment team should conduct focused interviews with knowledgeable representatives of healthcare providers, payers, healthcare information technology, interoperability, government, employers, researchers, and other appropriate participants.

The interviews need not be long. Thirty to sixty minutes should be sufficient if participants have considered the questions in advance.

Four questions should guide the discussion:

1. **What are the three to five most important problems, missed opportunities, or areas of significant effort in your part of healthcare?**

For each item described, ask:

2. **What information, knowledge, or capability do you wish you had that would help address it?**
3. **If that were available, what could you do differently?**
4. **What measurable value could result?**

The interviewer should probe for specific examples without steering the participant toward the proposed architecture. If the participant cannot estimate the overall value, ask for a specific example, the value associated with that instance, and how often similar situations occur. These examples can provide both a rough estimate of potential value and concrete evidence for the final assessment.

The objective is to establish a chain from:

Problem / Opportunity / Effort → Needed Information or Capability → Different Action → Measurable Value

Different participants will define value differently. It might include reduced administrative effort, faster decisions, fewer repeated tests, better treatment choices, reduced payment errors, shorter authorization times, better outcomes, improved patient experience, or opportunities not anticipated in this book.

Consolidate What Healthcare Says It Needs

The interviews should then be consolidated across organizations and disciplines.

AI could help analyze interview transcripts, identify recurring problems and desired capabilities, group similar opportunities, and find relationships that might otherwise be missed. The assessment team—not AI—would determine the meaning and significance of the findings.

For each significant opportunity, the team should document:

- the problem, missed opportunity, or effort being addressed;
- the information or capability participants said they need;
- what could be done differently if it were available;
- who would benefit;
- how the benefit could be measured; and
- the credible range of potential value.

This creates an evidence-based description of what healthcare participants say they could do with better information, independent of the architecture proposed in this book.

Then Compare the Needs With What Already Exists

Only after identifying the opportunities should the assessment turn to architecture.

The team should determine which healthcare information systems, TEFCA and QHINs, FHIR, IHE profiles, other interoperability initiatives, commercial products, or existing government and industry infrastructure already provide required capabilities.

The objective should not be to justify building something new. **If an existing capability already solves part of the problem, use it.**

The analysis should identify:

Needed Capability → What Exists Today → What Is Missing → What, If Anything, Must Be Added

This step may confirm parts of the architecture proposed in this book, modify them, or demonstrate that some proposed capabilities are unnecessary.

Assess Technical Feasibility

The remaining gaps can then be evaluated technically.

Could participating systems reliably communicate relevant new, changed, and deleted information? Can information describing the same patient and clinical events be matched across organizations? Can AI extract clinical meaning while preserving the underlying evidence? Can duplicate and conflicting information be identified and reconciled without concealing uncertainty? Can provenance and temporal history be preserved? Can relevant longitudinal knowledge be retrieved quickly enough to support healthcare decisions?

The assessment does not need to design the system. It needs to determine whether credible approaches exist for the major technical requirements and identify requirements for which feasibility remains uncertain.

Determine Whether the Potential Value Is Significant

Technical feasibility alone is insufficient.

The assessment should estimate the potential value of the opportunities identified through the interviews. Precision is neither possible nor necessary in 60 days. The objective is to determine whether credible benefits appear large enough to justify additional work.

Some benefits may be directly economic: administrative labor, duplicate services, payment errors, prior-authorization effort, or avoidable utilization.

Others may require different measures: physician preparation time, decision latency, patient burden, treatment effectiveness, safety, continuity of care, or improved ability to compare treatments and providers.

The important question is not whether every proposed benefit can be quantified. It is whether the combined evidence suggests an opportunity substantial enough to warrant further investigation.

Challenge the Findings

Before reaching a conclusion, the team should make the strongest case **against** proceeding.

The assessment should examine privacy, security, patient consent, governance, legal authority, patient identification, data quality, operational reliability, adoption, incentives, cost, concentration of risk, and other significant concerns identified during the work.

Classify each concern as a fundamental barrier, an unresolved question requiring further work, or a risk that could reasonably be addressed through architecture, governance, legislation, auditing, or implementation practices.

The purpose is not to defend the proposal. It is to determine whether it survives serious scrutiny.

The 60-Day Decision

At the end of 60 days, the assessment should answer the questions with which it began:

Is the potential value significant?

Is the required architecture technically feasible?

Are the major barriers reasonably addressable?

The result does not need to be simply yes or no.

It may conclude

1. **Stop** — The potential value does not justify further work.
2. **Resolve** — The opportunities appear promising, but specific questions must be resolved before determining whether to proceed.
3. **Proceed** — The potential benefits and technical feasibility justify a more detailed architectural and economic assessment.

In none of those cases does the 60-day assessment commit the government to a pilot, procurement, implementation program, or particular architecture.

Its purpose is much simpler: to determine whether the opportunity is important enough—and credible enough—to take the next step.

A Decision, not a Commitment

Healthcare has no shortage of committees studying broad problems.

This proposal calls for something smaller and more disciplined: six qualified people, 60 days, and a specific question.

Can a secure national longitudinal health information architecture give providers timely access to relevant patient history, reduce administrative friction, and create measurable benefits for patients, providers, and payers?

If the answer is no, document the reasons.

If the answer is yes, the nation can then decide what—if anything—to do next.

About the Author

Doug Ebel has spent much of his career working at the intersection of information, analytics, technology, and business decision-making.

During twenty-two years with Procter & Gamble, he developed information systems well before many of their underlying technologies became commonplace. In 1975, he built an enterprise data warehouse to support the integration of Folgers into P&G. In 1982, he designed and led development of a global electronic messaging and document-sharing system supporting 45,000 users worldwide. In 1984, he implemented Sales Force Automation using laptops. In 1988, he built a P&G enterprise data warehouse supporting the reorganization of sales into multifunctional customer teams.

His later work at Teradata included developing a patented Data Warehouse methodology and helping more than 100 corporations understand and evaluate advanced data and analytics technologies. He created the ClearScape Analytics Experience, a cloud-based environment that allowed tens of thousands of prospects, developers, and students to explore analytics and AI capabilities directly and for free rather than relying solely on presentations and demonstrations.

Across both P&G and Teradata, much of his work centered on a recurring question:

Can an organization get the right information to the right person in time to make a better decision?

After retiring and experiencing healthcare increasingly from the patient's perspective, Ebel began applying the same information-architecture questions to healthcare. Repeated medical histories, fragmented records, prior-authorization delays, and clinicians working without convenient access to complete longitudinal information led to the proposal developed in this book.

He is not a physician, and this book does not propose replacing clinical judgment with technology. It examines a different question: **Could better information architecture give patients, clinicians, payers, and other healthcare participants better information for the decisions they already must make?**

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The author retains responsibility for the architecture, arguments, conclusions, and final editorial decisions.

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